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Research paper

Topic: Geospatial evaluation of solar potential for hydrogen production site suitability: GIS-MCDA approach for off-grid and utility or large-scale systems over Niger

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ABSTRACT

Climate change mitigation requires environmental friendly and flexible alternative to fossil fuels such as green hydrogen with its renewable nature. The untapped huge solar radiation of Niger represents an opportunity to investigate green hydrogen production potential. Hence, this study proposes a novel geospatial multi-criteria decision analysis (MCDA) framework to perform site suitability analysis of solar photovoltaic (PV) based hydrogen potential for both off-grid and utility or large-scale systems over Niger. This research methodology combined land eligibility using the Geospatial Land Availability for Energy System (GLAES) tool and multi-criteria-based site suitability approaches involving environmental, socio-economic and technical criteria weighting applying an Analytical Hierarchical Process (AHP). The results show that the land eligibility analysis performed using GLAES demonstrates that 67.89 % (889,834 square kilometer) of the Niger territory is deemed eligible, while 32.11 % (420,738 square kilometer) represents the excluded constraints. The land suitability analysis for off-grid system shows that 32.36 % (421,842 square kilometer) has low suitability, 2.19 % (28,621 square kilometer) is considered moderately suitable, 26.1 % (340,184 square kilometer) of the eligible area has a strong suitability and 7.05 % (91,920 square kilometer) represents the most suitable area for hydrogen production. Similarly, land suitability analysis was performed for Grid-connected and large-scale system solar power plants to evaluate hydrogen potential. The framework of this study can be used as a spatial decision support system (SDSS) for policymakers and stakeholders to develop hydrogen roadmaps and strategies.

1. Introduction

The world economy is highly reliant on energy. Energy consumption is one of the critical indicators of a country's development. The world energy demand is increasing due to economic and population growth (IEA, 2024). By 2050, global energy demand is expected to grow by approximately 50 % compared to the levels recorded in 2020 (US Energy Information Administration (EIA), 2021). Additionally, most of the world's energy demand is met by fossil fuels despite their depleting nature and significant environmental impact, which has raised global concerns (Yang et al., 2021). Therefore, in 2015, the Paris Climate Agreement was endorsed by 195 nations to cap the rise in global temperatures to less than two degrees Celsius above pre-industrial levels. In recent years, there has been significant expansion in renewable energy driven by declining solar and wind power costs. According to the International Energy Agency (IEA), by 2028, renewable energy sources would contribute to more than 42 % of global electricity generation, with wind and solar photovoltaic (PV) energy doubling their share to reach 25 % (IEA, 2023). Ensuring access to sustainable and cost-effective energy while mitigating greenhouse gas emissions highlights the importance of exploring renewable energy options worldwide (Mohsin et al., 2018). The African continent has tremendous renewable energy

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(RE) potential, harnessing that could effectively change the continent's overall socio-economic condition (IRENA, 2021). For instance, access to electricity presents a significant challenge in Niger, with only 18.6 % of the population having access to electricity in 2021 (World bank, 2021). Thus, the national strategic plan target is to reach 100 % electricity access by 2035 (Ministère de l'Energie, 2018), although it seems quite ambitious regarding the current electricity access rate.

However, one of the significant drawbacks of renewable energy systems has been their intermittency, causing an imbalance in the supply-demand. Therefore, the challenge is to adjust the supply to meet the fluctuating demand precisely. Hence, storage is crucial to building innovative energy systems for a sustainable future (Mohsin et al., 2018). Promoting solar-powered hydrogen production can enhance electricity access by stabilizing the grid and ensuring reliable power supply, especially for utility-scale systems. Green hydrogen can also provide energy to remote areas where grid extension is impractical. As the world transitions to low-carbon economies, hydrogen, particularly green hydrogen, will play a crucial role. Developing countries aim to become a global energy leader, as outlined in the AU's 2063 Agenda. To achieve this, Africa can leverage its abundant renewable energy resources to produce green hydrogen to power various sectors. Consequently, the Africa-EU Energy Partnership and the 2019 EU Green Deal recognize hydrogen as a key component of a green transition and energy access strategy for Africa (AEEP, 2020). These developments necessitate a global assessment of hydrogen potential and the identification of suitable production and import locations. Though green hydrogen shows promise for a net-zero future, its production is constrained by limited resources like land and water (Tonelli et al., 2023). While several studies have used geospatial tools to assess the suitability of green hydrogen production sites, they often overlook critical factors such as proximity to water sources and the impact of dust, particularly in arid regions. Additionally, the selection criterion used in these studies are often generic and not tailored to the specific type of renewable energy system. This gap highlights the need for a more comprehensive and nuanced approach to site selection for green hydrogen production.

This study innovates by incorporating the geographical distribution of water resources and the impact of dust into the land suitability analysis, a factor often overlooked in previous research. It combines the GLAES tool with MCDA-AHP to assess both off-grid and utility-scale solar hydrogen production site suitability assessment for green hydrogen production. The study considers relevant criteria for techno-economic, social, and environmental feasibility. Given the global interest in green hydrogen, particularly the recent Germany-Niger partnership (Bhandari, 2022), this multidisciplinary approach offers a comprehensive framework for identifying suitable hydrogen production sites.

2. State-of-the art

This section examines the state of knowledge regarding green hydrogen production from three perspectives. First, it discusses the geographical factors influencing green hydrogen production. Second, it explores the land eligibility for hydrogen production. Third, it analyses the suitability of land for green hydrogen production using different criteria. Lastly, it compares the impact of these criteria on site selection and highlights the knowledge gaps in these areas.

2.1. Land eligibility

Renewable energy (RE) potential assessment is fundamental for achieving the world energy transition. Renewable energy production has significant geographical implications, particularly solar energy source. This implies that selecting a suitable site for solar energy power plant requires considering certain fundamental conditions the chosen site must fulfil (Asare-Addo, 2022) mostly optimal hydrogen production. The regional characteristics of the renewable energy sources underpinning green hydrogen production are a key factor influencing its costs and potential (Franzmann et al., 2023). Thus, the favorable renewable energy sources (RES) potential for green hydrogen production might be limited due to various constraints conditions and other types of land use (IRENA, 2022); making the use of Geographic information system (GIS) gaining interest in green hydrogen production siting. Considering the complexity involved in spatial planning for RE and hydrogen production infrastructure, utilizing GIS is essential (Müller et al., 2023). A preliminary requirement for assessing the costs and potential of green hydrogen is determining the availability and suitability of production sites. To comprehend land eligibility (LE) analysis, it's helpful to understand land use suitability analysis. This technique, used in various fields, identifies "the most appropriate spatial pattern for future land uses according to specify requirements, preferences, or predictors of some activity" (Malczewski, 2004). Land eligibility of a particular renewable energy source (RES) technology is defined as the binary conclusion dictating whether the technology in question can be placed at a particular location or no (Ryberg et al., 2017), considering a set of constraints that lead to the location being ineligible. When conducting a land suitability analysis, it's crucial to distinguish between site selection problem and site search problem. Site selection involves ranking a predefined set of potential sites to identify the optimal choice. In contrast, site search aims to establish the boundaries of sites without a pre-determined set of candidates (Malczewski, 2004).

In the context of renewable energy (RE), site selection or assessment methodologies typically involve four stages. The initial stage, exclusion, involves eliminating areas that don't meet specific criteria, known as exclusion criteria (EC). These criteria often involve thresholds or acceptable ranges. The second stage focuses on selecting assessment criteria (AC) to evaluate the suitability of potential sites. The third stage assesses these potential sites against the defined AC. Finally, the fourth stage optimizes the assessment results to identify the most suitable site (Spyridonidou and Vagiona, 2023). The first stage is equivalent to the site search problem referred to the LE or LE analysis. The three remaining steps corresponds to the land or site suitability analysis, which is explained in the subsequent part of the work. This study involves both land suitability analysis and land eligibility analysis.

LE analysis usually undertaken on a regional scale, the analysis is constrained by the geographical context and the type of renewable energy technology involved (Ryberg et al., 2017). The reviewed studies in (Spyridonidou and Vagiona, 2023) on the LE analysis for green hydrogen production using RES at the country scale shows that the LE analysis rely on local land use regulations and policies. Similarly, from the reviewed literature by Ryberg (Ryberg et al., 2017) on LE analysis for various RES identified significant inconsistency in three aspects: Criteria definitions, LE methodologies, and data use. Mostly in developing countries there is a lack of clear policy regulation for the criteria selection encompassing physical, environmental and socio-political constraints for RE potential evaluation particularly for green hydrogen production. This study offers policy and regulatory insights into the criteria that influence the geographical suitability of solar energy production for green hydrogen in Niger. For instance,a geospatial analysis is conducted to evaluate the potential for green hydrogen production in Togo using biomass, wind, and solar energy. While the exclusion criteria prioritize environmental aspects, they overlook important social and techno-economic factors like infrastructure proximity and land use for various purposes. Furthermore, the scale of renewable energy technology implementation is not clearly defined (Simon et al., 2022). Therefore, to overcome the LE analysis inconsistency, a land eligibility framework was developed by Ryberg (Ryberg, 2019) called Geospatial Land Availability for Energy System (GLAES). The methodology developed in GLAES can operate in any geographical context and accommodate the most common geospatial data formats. This model aims to promote the standardisation of future LE applications regardless of how the criteria are defined and the datasets are used. The model can be found on GitHub under the project name Geospatial Land Availability for Energy Systems (Ryberg et al., 2018b) https://github.com/FZJ-IEK3-VSA/glaes. Initially, the GLAES is applied in many study focus on evaluation LE for RES (Caglayan et al., 2019; Peña Sánchez et al., 2021; Ryberg et al., 2018b; Tlili et al., 2020). A large and growing body of literature has investigated RES site selection using land eligibility methods. Land eligibility has also been employed to construct inputs for energy system models in the context of Germany (Ryberg et al., 2018b) and in UK (Samsatli et al., 2016).

One of the first application of land eligibility analysis using GLAES is carried out by Ryberg over Europe. In the LE analysis 36 commons constraints have been used to evaluate the eligible area for RES. A sensitivity analysis on the different constraints shows the effect of the excluded constraints overlay. The study concluded that the GLAES model can apply for other countries for land eligibility analysis considering the context-specific exclusion constraints selection and a starting point of eligibility analysis methods standardization among researchers (Ryberg et al.,2018b). Ryberg has concluded that the impact of eligibility constraints applied is highly spatially sensitive and that constraints based on woodland, agriculture areas, irradiance, and protected areas are more influential in the analyses (Ryberg et al., 2020). Techno-economic assessment of variable renewable energy sources in Mexico, considering the temporality and geospatial constraints was conducted by Sanchez (Peña Sánchez et al., 2021). In the study the land eligibility analysis is performed using the GLAES tool, for wind and solar Photovoltaic (PV) respectively using onshore, offshore wind turbines and open-field, rooftop photovoltaic. A country-specific factors are considered for the constraints and the buffer distance thresholds. Seven primary context specific constraints have been added for the LE analysis namely, military areas, harbors, Liquefied Natural Gas (LNG) terminals, geothermal sites, primary jungles, and active volcanoes and hurricanes. The results shows that 91 % of the electricity would be generated from open-filed PV representing 578,000 square-kilometer while only 9 % of the electricity would be generated from onshore wind corresponding to 68,500 square kilometer of the eligible area, offshore and rooftop PV contribute to less than 0.03 % of the electricitygenerated in Mexico (Peña Sánchez et al., 2021). For example, Lopez investigated land eligibility analysis using a portfolio of technology in the United States, including open field PV, concentrated solar power (CSP), and onshore wind. Multiple classifications of the protected areas, human settlement, orographic characteristics of the terrain, socio-political criteria, and water bodies have been considered as constraints (Lopez et al., 2012).

Furthermore, land eligibility investigation for green hydrogen is carried both for local and regional scale. In Germany, eligibility analysis is performed for spatio-temporal of a future energy system for Power-to-Hydrogen application (Welder et al., 2018). To decarbonize the Spanish transport sector by 2050, a hydrogen supply chain analysis is conducted. The study, firstly analyse the eligible area using the GLAES framework. The land eligibility analysis results show that 29.14 % and 18.22 % of the area is deemed eligible for respectively eligible for solar PV and onshore wind systems out of the 493,549 square-kilometer of the regions territory. The study conducted by Olfa to evaluated the French electricity generation capacity to the regional scale that includes LE analysis using GLAES and Multicriteria analysis for the geospatial site suitability investigation. The criteria for the site suitability analysis selected are mainly based on the techno-economic considerations, while a linear weighting of the criteria on a scale from zero to one is applied accordingly to the distance or renewable energy resources (Tlili et al., 2020). Although, the GLAES tool is versatile and applicable to diverse regions and datasets, its application in research is limited in sub-Saharan Africa. This study addresses this gap by focusing on sub-Saharan Africa, specifically Niger, to develop a standardized set of exclusion criteria for green hydrogen production land eligibility, informed by literature review and expert insights. While it is widely acknowledged that land suitability analysis involves excluding unsuitable areas through eligibility analysis, the majority of the reviewed studies focus on identifying eligible areas without providing a more detailed analysis, such as ranking or comparison, to assist decision-makers in selecting the most suitable sites. The following section will present a review of studies on land suitability for green hydrogen production, summarizing relevant literature on the topic.

2.2. Land suitability analysis

Spatial decision problems are characterized by a wide range of alternative and a complex set of conflicting and incommensurable evaluation criteria (Malczewski, 2007). Many spatial decision problems can be effectively addressed by combining Geographic Information Systems (GIS) with Multicriteria Decision Analysis (MCDA). These two fields complement each other well. GIS provides powerful tools for analysing spatial data, while MCDA offers methods for structuring complex decisions, evaluating options, and prioritizing choices. By integrating these two approaches, GIS-MCDA can help decision-makers make informed choices in a variety of spatial planning and management contexts (Chakhar & Martel, 2003; Laaribi, Chevallier, and Martel, 1996; Malczewski, 2006). Recent years have seen significant research into GIS-MCDA techniques, which are effective tools for identifying optimal locations for renewable energy source in diverse climatic and geographical settings (Kumar et al., 2017a). Given the challenge of locating suitable sites for cost-effective green hydrogen production, GIS-MCDA can be a valuable tool. By combining GIS and MCDA, a spatial decision support system (SDSS) can aid policymakers and stakeholders in making informed decisions, especially when faced with multiple and conflicting objectives (Malczewski et al., 2015). For instance, Kumar's review highlights the use of MCDA in energy planning, focusing on key indicators that can help developing nations achieve sustainability goals across various energy systems. The different key indicators of the reviewed articles have been highlighted for various energy systems schemes which can be utilised to address the core issues for achieving the goal of sustainability in developing nations (Kumar et al., 2017a).

Recently, various methodologies for assigning weights to renewable energy criteria were investigated. For instance, Choi has carried out review on GIS-based solar radiation mapping, site suitability and potential assessment, aiming to elucidate the role of GIS as a problem-solving tool for PV and Concentrated Solar Power (CSP) systems for electricity generation (Choi et al., 2019). Some methods mentioned include the Preference Ranking Organisation Method for Enrichment Evaluations (PROMETHEE) by Brans and Vincke (Brans and Vincke, 1985), the Technique for Order Preference by Similarity to Ideal Solutions (TOPSIS) by Huang and Yoon (Huang and Yoon, 1981), Elimination and Choice Translating Reality (ELECTRE) by Benayoun (Benayoun,R et al., 1966), Weighted Product Model by Bridgman (Bridgman, 1922), Boolean Overlay by Jensen (Jensen and Christensen, 1986), Weighted Average by Anderson (Anderson, 1967), Weighted Sum Model by Fishburn (Fishburn, 1967), Fuzzy Analytic Hierarchy Process (FAHP) by Zadeh (Zadeh, 1965) and the Analytic Hierarchy Process (AHP) by Saaty (Saaty, 1980), etc. These methods have been applied in various studies in many countries, such as Sudan (Zalhaf et al., 2022), China (Sun et al., 2013), Serbia (Doljak et al., 2021), Canada (Senteles, 2017), USA (Janke, 2010), Spain (Díaz-Cuevas et al., 2019), Indonesia (Ruiz et al., 2020), Greece (Latinopoulos & Kechagia, 2015), Ireland (Piirisaar, 2019), Turkey (Atici et al., 2015), Saudia Arabia (Al Garni and Awasthi, 2017), Palestine (Braik et al., 2022), Ethiopia (Gerbo et al., 2022), Mauritius (Doorga et al., 2019), in West Africa (Yushchenko et al., 2017), in Ghana (Asare-Addo, 2022). Although numerous Multi-criteria Decision-Making (MCDM) techniques are available, the one that finds the most extensive application is the AHP technique, pioneered by Thomas Saaty (Saaty, 1980). Nevertheless, while the AHP method proves valuable, it should be noted that it lacks comprehensiveness when employed in isolation (Kumar et al., 2017a). To illustrate, the AHP approach does not consider geographical conditions at a given location. Consequently, several studies have opted to combine the AHP methodology with other techniques like fuzzy logic and Goal Programming, among others, to simplify decision-making, especially when dealing with many criteria (Ramanathan and Ganesh, 1995). One of the widely weighting criteria used is the AHP approach combined with GIS to identify optimal locations for renewable energy projects. These studies, conducted in various countries like Mauritius, Mozambique, Saudi Arabia, Iran, and Indonesia, consider factors such as environmental, economic, and technical criteria (Albraheem and Alabdulkarim, 2021a; Jay et al., 2018; Tafula et al., 2023a). While there's a study by Laouali focused on solar potential in Niger using GIS, it doesn't delve into detailed site selection using MCDA techniques (Laouali et al., 2019).

In recent years, researchers are gaining interest in applying the GIS-MCDA and AHP weighting technique for green hydrogen potential site suitability evaluation. For instance, in Argentina, Sigal assessed the potential for hydrogen production from renewable resources. The study evaluated the geographical and technical potential of hydrogen production from wind, solar PV, and biomass energy (Sigal et al., 2014) by considering some critical socio-economic and environmental criteria to find out the site suitability. In contrast, Razzaqul, in the Sultanate of Oman study, priority has been given to the economic analysis of solar-to-hydrogen production (Ahshan, 2021). The study ranks the economic suitability analysis of hydrogen production without including the geographical constraints for hydrogen production, such as the proximity to infrastructure that could increase the cost of production. A thorough analysis of the development of renewable (solar PV and wind) hydrogen production projects is carried out in Uzbekistan. The multi-criteria decision-making was applied based on the various technical potential criteria, and an economic analysis was performed (Mostafaeipour et al., 2021). In the study, the multi-criteria method has yet to be combined with the GIS to determine the relationship between the potential and the related location. For instance, Messaoudi studied potential site selection for solar hydrogen production using GIS-based multi-criteria decision-making in Algeria. The study focuses on site suitability analysis and technical potential using AHP approach for criteria weighting; therefore, interest has yet to be given to the type of solar-based hydrogen system (Messaoudi et al., 2019). Similarly, a study conducted by Samir on the techno-economical assessment of the hydrogen potential. The study used the geospatial approach to determine the green hydrogen potential; the land constraints have not been considered in the analysis (Touili et al., 2018). This study is limited by the lack on considering water resources in both geographic and techno-economic feasibility analysis although the key relevance in the green hydrogen production. In 2022, the International Energy Agency (IEA) assessed the cost and potential of green hydrogen production using solar and wind energy to align with the 1.5°C climate goal. This global assessment, conducted at a 1×1 km^2 resolution, considered factors like land eligibility and techno-economic feasibility (IEA, 2022), However, one of the key limitations of the study is overlooking the geographical implication on the cost of green hydrogen at the region level. Stepanov's research on utility-based hydrogen production siting indicates that locations closer to power lines are more suitable for hydrogen production due to lower energy transmission costs (Stepanov et al.,2016). Table 1 summarizes a comparative analysis of the research studies. Additionally, the African Development Bank Group stresses the significance of proximity to freshwater sources for green hydrogen production, suggesting sites near rivers and lakes (Afdb, 2022). Emmanuel further emphasize the importance of road proximity, highlighting the use of GIS to identify sites with better road connectivity, potentially reducing transportation costs and improving logistics for green hydrogen production (Emmanuel, 2022). The importance of proximity to infrastructure varies

Table 1
Literature review.

Publication	Scope of the study area	Water Bodies	Agricultural Land	Settlements	Road Network	Slope	Protected Area	Important Birds Area	Small Airports	Medium and Large Airports	Proximity distance to the Road	Proximity Distance to the grid	Proximity to the hydrogen demand	Slope reclassification	Renewables energy source	Pipeline	Desalinated Water source	Dust distribution	Proximity to the Water sources	Power system consideration	Hydrogen production	Suitability analysis evaluation
		Land eligibility Analysis Constraints									Site Suitability analysis criterion											
(Messaoudi et al., 2019)	Algeria		x																	x	x	
(Ishmam et al., 2024)	Benin	x	x	x	x		x	x	x	x					x		x		x			
(Rahmouni et al., 2017)	Algeria														x				x		x	
(Sigal et al., 2014)	Argentina	x	x	x		x	x	x					x		x						x	
(Romero Boix, 2023)	Spain	X	X	X	X		X	X	X	X	X	X	X		X		X		X		X	X
(Ali et al. 2022)	Thailand				x	x	x						x	x	x				x		x	x
(Asare-Addo, 2023)	Ghana						x	x				x	x		x					x	x	x
(Simon et al. 2022)	Togo						x	x							x						x	
(Touili et al. 2018)	Morocco														x						x	
(Ryberg et al., 2018b)	Europe	x	x	x	x	x	x	x	x	x	x	x										
(Winkler et al., 2024)	SSA	x	x	x	x	x	x	x	x	x	x	x	x		x		x		x	x	x	

Ishmam et al., 2024; Rahmouni et al., 2017; Romero Boix, 2023; Winkler et al., 2024 depending on the type of hydrogen production technology. For example, while proximity to the power grid may be less critical for off-grid or large-scale export-oriented hydrogen plants, it is essential for utility-scale hydrogen plants. The aforementioned literature shows that none of the reviewed studies that used GIS-MCDA considered the impact of dust, particularly in sandy areas, as a factor in ranking site suitability (Alhammad et al., 2022; Almutairi et al., 2021; Haddad et al., 2020; Messaoudi et al., 2019) although the effect on the panel efficiency and electrolysis system. For instance the result of the study carried out by Danu shows that more than 14 g m^{-2} causes a poly-Si module to lose more than 30 % of its efficiency (Danu et al., 2018). Dida et al. found that a 30 MWp PV power plant in Algeria experienced a 32 % energy reduction after a sandstorm incident, highlighting the impact of Saharan environments on crystalline silicon modules (Dida et al., 2020). To further investigate the effect of dust on photovoltaic module performance, Majeed and colleagues conducted outdoor experiments in Pakistan (Majeed et al., 2020). . They observed a significant decline in power output for both mono-crystalline silicon and polycrystalline silicon modules after one month of exposure to dust. This decrease, amounting to 16.16 % and 11.54 % respectively, correlated with a dust deposition density of 4.6 g per square meter (Majeed et al., 2020). This study represents one of the initial endeavors to account for dust distribution density when selecting optimal locations for solar hydrogen production systems. The reviewed literature revealed three primary limitations:

- A notable gap in existing research regarding the geospatial factors influencing green hydrogen production site suitability in Niger,
- The oversight of solar-based green hydrogen production system types in the selection of suitability analysis criteria,
- The lack of consideration for the specific context of the study area in the selection and weighting of criteria.

However, this study addresses a significant research gap by investigating the geospatial implications of green hydrogen production site suitability analysis in Niger. A novel aspect of this research is the incorporation of local context and expert insights into the selection of criteria. By considering the relevance of water sources and dusty areas, this study provides a more comprehensive and tailored approach to site suitability assessment. Furthermore, the findings of this study have important policy implications, as they can inform decision-making on the optimal location of green hydrogen production facilities in Niger, maximizing both environmental and economic benefits.

This study is structured as follows. First, we conducted a comprehensive literature review to identify research gaps in geospatial assessments of solar-based hydrogen potential and summarized the state-of-the-art findings. Next, we outlined the methodology for land eligibility analysis, including the criteria used for land suitability assessment and the applied weighting methods. Finally, we presented the results of land eligibility analysis and the potential for hydrogen production site suitability for off-grid, grid-connected, and large-scale systems. Additionally, we evaluated land suitability for large-scale solar-based hydrogen systems, considering domestic water resources (underground and surface water) and the potential for importing desalinated water from Benin and Nigeria.

3. Methodology and data

3.1. Study area

Niger is a landlocked West African country characterised by particularly harsh climatic conditions, surrounded by Chad in the east, Nigeria and Bénin in the south, Mali, Burkina Faso in the west and Libya and Algeria in the north. Two-thirds of its land area lies within the Saharan zone, while the remaining one-third is situated within the Sudanian and Sahelian zones (Niger NDCs, 2021). The country is heavily affected by climatic uncertainties, experiencing significant interannual, spatial, and temporal variability in rainfall. Depending on the season, daily maximum temperatures range between 31 and 42 degrees Celsius (World data, 2021). The estimated population of Niger stands at 27,640, 265, with a population density of 21 inhabitants per square kilometre, across a total land area of 1267,000 square kilometer (INS, 2023). Electricity access remains a challenge in Niger, with a significant share of the electricity consumption being imported. Therefore, the huge untapped solar potential with a global horizontal irradiation range between 5.77 – 6.53 kWh/m^2 averaging 6.265 kWh/m^2 represents an alternative for the country's electrification in decarbonising the power sector and transport to achieve the country's National Determined Contribution (World Bank, 2023). The Fig. 1 present the study area of the current study.

3.2. Methodology

This study is divided into two main parts, both focused on the geospatial analysis of solar-based hydrogen potential. The first part uses the Geospatial Land Availability for Energy Systems tool to perform the land eligibility analysis regarding the various economic, technical and environmental constraints. Moreover, the second part evaluates the site suitability analysis using the AHP-Multi-criteria methods. To fill the gap in the literature, the site's suitability was implemented based on the type of solar power plant. Subsequently, the criteria selection depended on the solar-based green hydrogen system type. Intense investigation from the literature and expert interviews are conducted for the suitability analysis criteria selection for off-grid and utility or large-scale solar hydrogen production systems. The large-scale solar-based hydrogen production site is evaluated for future hydrogen export using domestic water sources (surface and groundwater) and desalinated water importation. Subsequently, in the criteria definition, the consideration was given not only based on the power infrastructures (Ahmadi et al., 2022; Mushtaq et al., 2022), But also, on the transport infrastructure needed to supply hydrogen to the various shipping ports. Therefore, Sèmè (Benin) and Warri (Nigeria) ports are considered as two scenarios of hydrogen supply from Niger prior to the respectively Niger-Benin oleoduc path (Business Africa, 2021) and Trans-Saharan gas pipeline projects (Ahmed, 2022). A sensitivity analysis of the criterion is carried out following the equally weighting approach of the selected criterion according to the typology of the system. Additionally, most previous studies have neglected to consider factors like water supply sources, dusty areas, and pipeline infrastructure when assessing the suitability of sites for hydrogen production (Ackah et al., 2021; Ahmadi et al., 2022; Ahshan, 2021; Ishaq and Crawford, 2024; Messaoudi et al., 2019).

3.2.1. Land eligibility analysis

Land eligibility analysis for solar hydrogen is conducted using the Geospatial Land Availability for Energy Systems model developed on Python. GLAES is an open-source model aiming to standardised the implementation of LE analysis (Ryberg et al., 2018b). As presented in the GLAES framework Fig. 2, the study region must be defined using the vector layer of the administration boundary and initialising the analysis on the target country's spatial resolution and reference system. Vector layer rasterization is performed for the vector constraints layers to allow the exclusions of the ineligible area otherwise for the raster file. Multiples or single exclusions can be applied to select the eligible location in the region of interest by discarding the ineligible land. A buffer distance calculation is included to guarantee infrastructure safety such as pipelines, roads, power grids, ecosystem preservation (protected area land, important bird area, etc.), and socio-political legislation for the inhabitant's safety. In addition, the buffer distance to the target constraints is customised according to the local criteria and experts' insight as presented in Table 2. Several restriction constraint should be considered when installing solar-based hydrogen for a more socio-environmentally friendly system (Suprova et al., 2020). Delimited

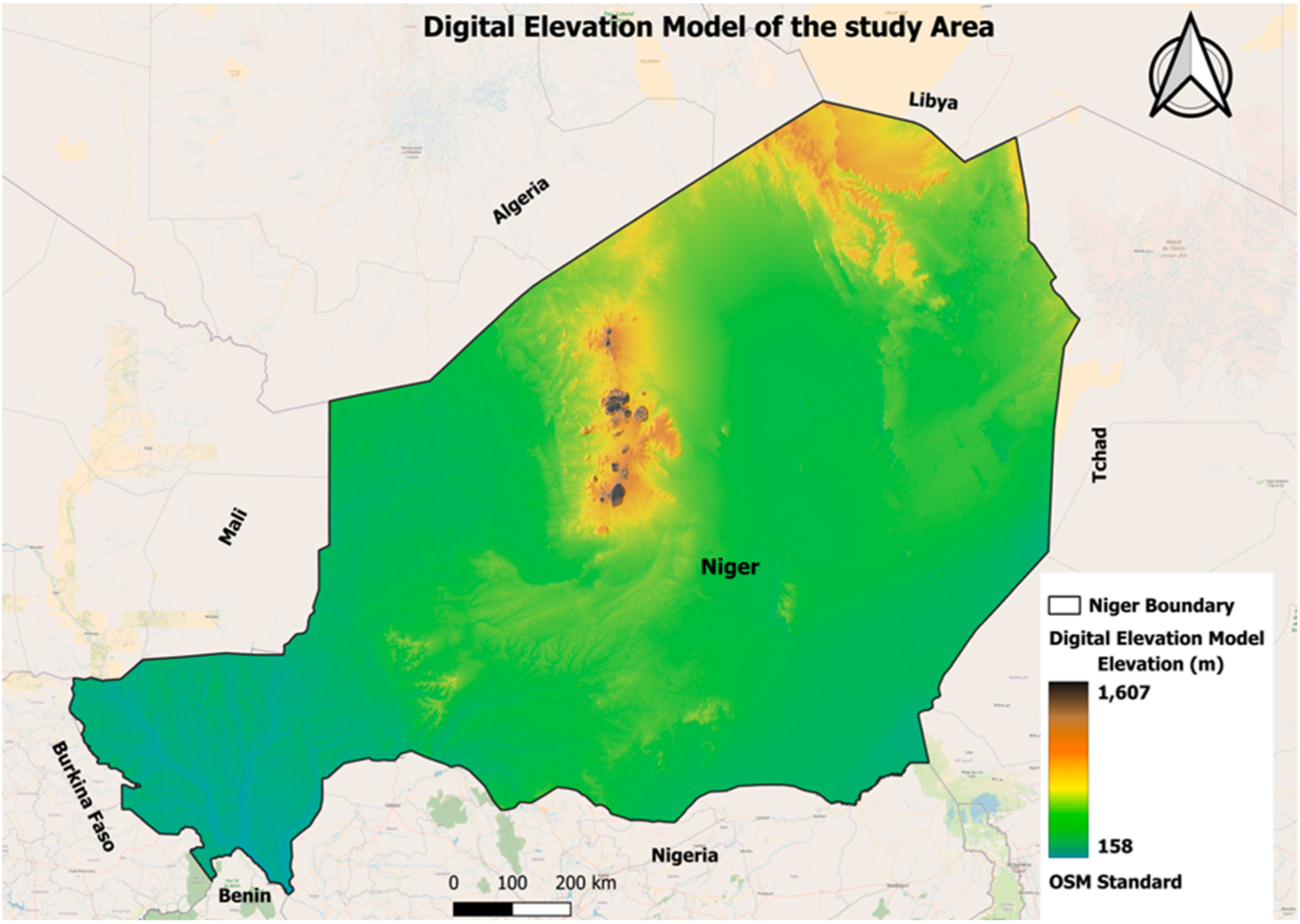

Fig. 1. Study area map.

areas like international protected areas are excluded from the eligible site selection mainly for ecological and environmental protection reasons. The protected areas are some of the significant restricted areas considered, such as forests, water sources, naturally protected areas, essential bird areas like Ramsar sites, forest reserves, parks, etc (Yushchenko et al., 2017). As a result, a 1000 m buffer distance was considered for the protected areas. The protected area datasets were collected from the open access source at the United Nations Environment Programme (UNEP-WCMC) and the International Union for Conservation of Nature (IUCN) (UNEP-WCMC & IUCN, 2023). The Table 2 provides the datasets of the exclusion constraints.

3.2.2. Land suitability analysis for solar-based hydrogen production

This study combined the geographical information system (GIS) with the multi-criteria decision analysis (MCDA) to identify hydrogen site suitability. In the analysis raster and vector layers data type are used for site solar-based hydrogen production site suitability analysis. The study considered nine criteria for site suitability analysis, regardless of the system type. Most of the data, such as information on roads, power grids, water sources, and pipelines are in vector format. However, solar radiation, slope, and dust density data were obtained in raster format. Given that most of the data was in vector format, the first step in the spatial analysis involved converting these vector layers into raster format using the rasterization tool in QGIS (QGIS, 2023). To ensure consistency and efficient processing, all raster layers were reprojected to the EPSG:3857 coordinate system and resampled to a 100 m x 100 m resolution using the resampling processing Toolbox in QGIS over Niger boundary layer. This standardization facilitates analysis and reduces computational demands. Afterwards, the distance to various infrastructure (roads, power grid, pipeline) and water sources was calculated using the raster proximity function. Surface and groundwater sources were combined into a single layer using raster calculator toolbox, representing domestic water sources. To ensure that the distance calculations are relevant to the study area, the distance layers were clipped to the study area boundary using(QGIS, 2023). The obtained distance layers were reclassified based on proximity to the target features using the reclassification toolbox. Moreover, continuous raster layers like solar irradiance, dust, and slope were directly reclassified into categories after pre-processing. The proximity distance of the selected criterion is reclassified as depicted in the Table 3. The raster layers are normalized for the overlay analysis using the suitability analysis criterion and the eligibility analysis layer. The output of the overlay analysis is process using the zonal statistic toolbox (Ali et al., 2022; Tafula et al., 2023a), as displayed in flowchart diagram of the Fig. 3. For instance, proximity to grid power is essential for grid-connected solar power systems. However, it is not considered for the off-grid solar-based hydrogen system or large-scale potential production site suitability. Conversely, off-grid site suitability for hydrogen production needs to be closer to the remote settlement; therefore, the site has to be further away from the settlements for utility or large-scale systems. Moreover, according to the type of system, the factors for site suitability analysis are grouped into two types of main criteria, either technical or economical, while the environmental factors are considered excluded constraints and discarded for the safety preservation of the environment, as in (Messaoudi et al., 2019). Respectively, factors highlighting the challenges related to the feasibility of solar-based hydrogen systems are considered technical

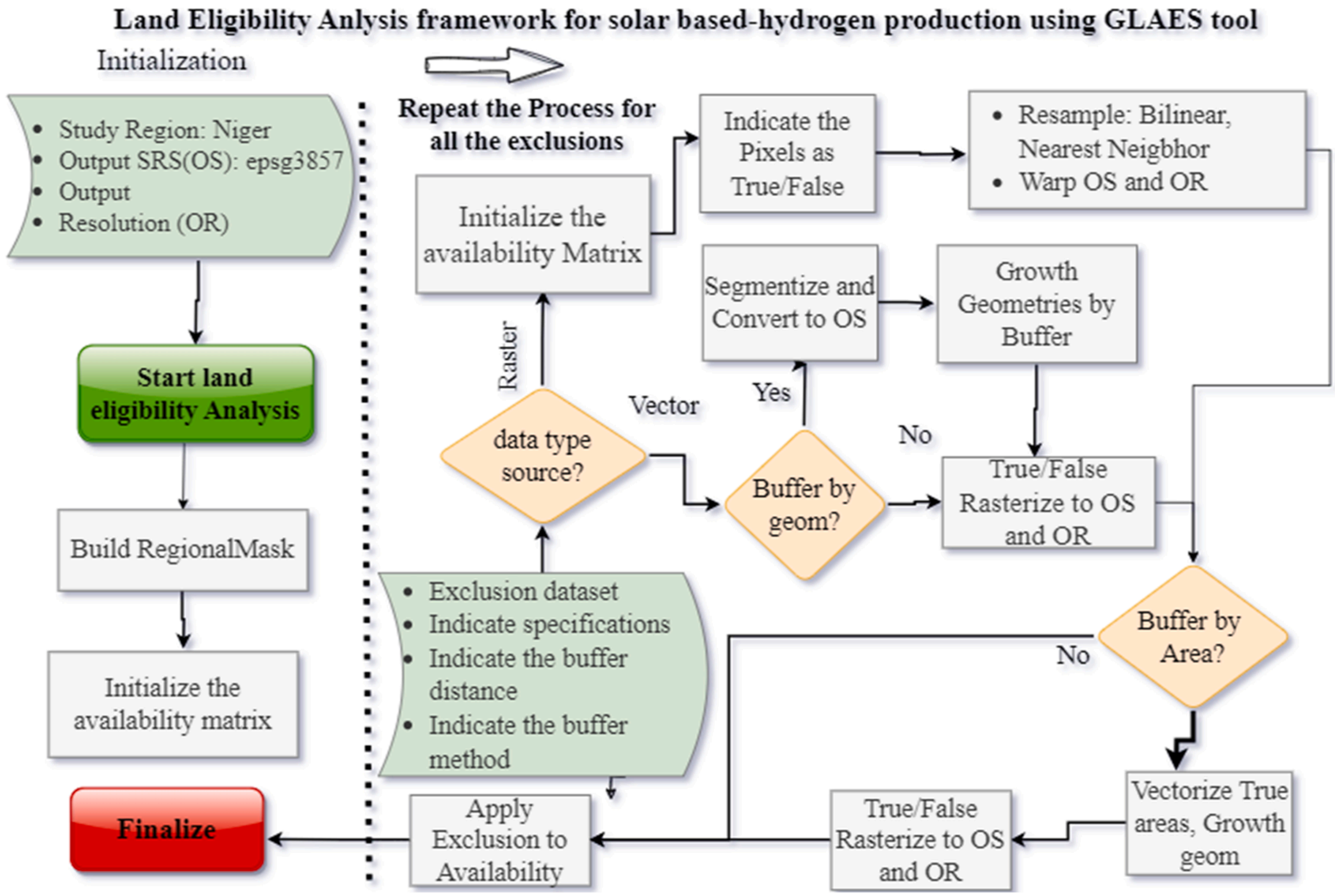

Fig. 2. Land eligibility analysis framework using the GLAES tool adapted from (Ryberg et al., 2018b)

criteria. Conversely, the economic criteria affect the cost of hydrogen production based on the proximity and distance of the related factors. Moreover, for the continuous raster, such as solar irradiance, dust, and slope after pre-processing, reclassification was directly realised for the categorisation of the data.

The developed methodology framework of Fig. 3 is followed to evaluate the suitability analysis of the hydrogen production site, which can be summarised in the following steps:

- Criteria Data collection and pre-processing according to the type of system,
- Spatial analysis, which includes several methods such as surface analysis, geometric operation and distance operations using the function developed on Python and QGIS tools,
- Then, the spatial decision support, whereby the multi-criteria decision analysis (MCDA) is applied using an analytic hierarchical process (AHP),
- Finally, a solar-based hydrogen site suitability (SHSS) is generated by giving each raster its weight and then combining them with the restriction using the union overlay method using the Eq. (1) (Albraheem and Alabdulkarim 2021b). The geographical potential result is extracted using Zonal statistic

$$SHSS = \sum_{i=1}^{n} x_i \times w_i \times r \tag{1}$$

Where r is the restriction constraints, x_i: criteria value layer and $w_i$is the weight of the criteria. The criteria for site suitability analysis for solar-based hydrogen production are presented in the subsequent section.

a) **Solar irradiance:**

The primary requirement for initiating a solar PV project is ensuring the availability of resources. It is crucial to ascertain whether the chosen location receives sufficient solar irradiance suitable for generating electricity and producing green hydrogen. This criterion is a critical factor in assessing the suitability for implementing solar technology (Asare-Addo, 2022; Suprova et al., 2020). Global horizontal irradiance (GHI) is considered the main input criterion for evaluating site suitability, and it was collected from the solarGIS platform (World Bank, 2023). The solar irradiance data was reprojected from EPSG 4326 to EPSG 3857 and resampled to a 100-meter resolution using bilinear interpolation. The GHI reclassified layer is presented in the Fig. 4. The choice of solar irradiation weighting differs considerably among the studies due to the challenge related to resource availability. In Niger, due to the abundant solar potential, importance was given to the economic factors for reducing the cost of hydrogen production (Yushchenko et al., 2016). The normalized solar irradiation is used in the overlay analysis.

b) **Grid infrastructure:**

Solar-based hydrogen production for power systems is a promising solution to overcome the grid instability caused by the injection of variable renewable energy sources. The power plant's proximity to the electricity grid is necessary to reduce the transport cost of the electricity to the transmission grid. The proximity distance to the grid network is minimised for utility-scale systems to reduce the system's connection cost. Conversely, the distance to the grid is maximised for off-grid systems to avoid installation near the power grid. The power grid data, including both existing and planned transmission and distribution networks for the West African Power Pool are collected in vector format. After clipping the Africa-wide

Table 2
Land eligibility exclusion and suitability analysis criterion.

Exclusion criteria factor	Buffer distance (m)	Format	Spatial Resolution	Source
Forest	100	Vector	500 m	(Copernicus Global Land Services CGLS, 2022)
Natural Protected Area	1000	Vector	500 m	(UNEP-WCMC and IUCN, 2023)
Important Birds Protected Area	1000	Vector	500 m	(UNEP-WCMC and IUCN, 2023)
Waterbody (Surface & Ground)	300	Vector	500 m	(OSM, 2022)
Primary Roads Networks	200	Vector	500 m	(OSM, 2022)
Secondary roads networks	100	Vector	500 m	(OSM, 2022)
Power Grid	200	Vector	500 m	(World Bank., 2015)
Land use	200	Vector	500 m	(OSM, 2022)
Settlement	500	Raster	100 m	(World Bank, 2021)
Small Airport	1000	Vector	500 m	(OSM, 2022)
Large/medium airport	2000	Vector	500 m	(OSM, 2022)
Desalinated Water route	300	Vector	500 m	Georeferenced Map
Pipeline	300	Vector	500 m	Georeferenced Map
Dust density distribution	no	Raster	30 m	(Earth Data, 2023)
Global Horizontal Irradiance	no	Raster	100 m	(World Bank, 2019)
Slope	no	Raster	90 m	(CGIAR-CSI, 2008)

data to the Niger boundary, the existing and planned grid layers were merged into a single layer using the vector processing toolbox. The proximity to the grid is not considered suitable for exporting large-scale solar-based hydrogen production, the Fig. 5 depict the reclassification layer. The data on the transmission power grid was collected from the OpenStreet map (OSM) (World Bank, 2015) in vector layers.

c) **Road Network:**

This study considers both primary and secondary roads. Access to sites via road networks significantly reduces installation and maintenance costs for power plants (Ryberg et al., 2017). Road data was sourced from OpenStreetMap (OSM, 2022) and Natural Earth Geoportal, and the Quick Street tool was used to update the road network layers. Proximity to roads not only minimizes infrastructure costs during construction but also reduces maintenance costs post-installation, while also mitigating environmental impacts (Acar, 2018; Van Haaren and Fthenakis, 2011). The vector-based road data was converted into a raster format and resampled for analysis. Both primary and secondary roads were considered in the analysis, proximity distance reclassified shown in the Fig. 6.

d) **Pipeline infrastructure:**

Similarly to power lines, pipeline routes may significantly influence the site selection procedure (Soha and Hartmann, 2022). Natural gas pipelines could be used to transport the hydrogen produced from VREs by blending it with natural gas or repurposing the pipeline to green hydrogen transport (Jayanti, 2022), which could be a viable economical solution to transport and distribute hydrogen (Josmar et al., 2023). The pipeline geospatial data was created from scratch using the georeferencing toolbox and Google Earth to draw the pipeline route as a shapefile. In this case, the transaharian gasoeduc pipeline project (TSGP) (Ahmed, 2022) for transporting natural gas from Nigeria to Algeria by crossing Niger, proximity distance is explored for suitability analysis. Additionally, proximity distance near the West African Gas Pipeline Company (WAPCO) Niger-Benin oil pipeline is suitable as feasibility was carried out for oleoduc routing (Africa Business, 2021). The Fig. 7 shows the reclassified raster of the proximity distance to the pipeline for the spatial analysis.

e) **Water source:**

In electrolysis-based hydrogen production, water is a key input for converting electricity from variable renewable energy to hydrogen using electrolyser. As fuel is not needed, the main operational cost of water electrolyser-based hydrogen production is water consumption (Tang et al., 2022). Producing hydrogen using water as input in dry area is challengeable. Domestic water resources, including ground and surface water (river + lack), are selected for site suitability analysis for utility and off-grid systems. Therefore, the two type of water sources are merged to one single vector layer. The River and groundwater source data are collected respectively from Natural Earth Geoportal (Natural Earth, 2022) and the West African groundwater database (BGR et al., 2022). The obtained hydrological data over Niger territory is provides from the Africa Groundwater Atlas which mapped the hydrogeological unit overview of each ECOWAS countries at a scale of 1:5 million. The harmonisation of the geological unit is provides from various studies to investigate the Iullemenden, Tchad, and Djado Basins in Niger over a scale of 1:5 million and the Taoudeni basin, Inner Niger delta (British Geological Survey, 2019; Greigert and Bernert, 1978; Seguin, 2016). The aquifer expected productivity is classified into classes from high productivity aquifers (> 10 l/s) with a discharge capacity of 10 l/s a limited aquifer productivity (0.1–0.5 l/s) (BGR et al., 2022). For the water proximity distance evaluation, the high productivity aquifer which allow groundwater supply for large towns, large industries or large-scale irrigation projects is considered for Large-scale and utility scale solar based hydrogen production site suitability analysis. In contrast, for off-grid solar based green hydrogen production suitability sitting the moderate aquifer productivity (2–10 l/s) is used taking into consideration the sustainability of the water source for the various end-users and its renewability. Although Niger is located in the tropical zone, a recent study carried out in March 2023 by the millennium corporation challenge revealed that Niger is one of the richest countries with groundwater sources in the Sahel (BGR et al., 2022). Subsequently, this represents a massive opportunity for the

Table 3
Criterion reclassification.

Score	GHI (KWh/m2)	Proximity to road (Km)	Proximity to Grid (Km)	Slope (%)	Proximity to Waterbodies (Km)	Proximity to Sea water route (Km)	Proximity to Dust density Distribution (10–5 Kg/m2)	Proximity to Pipeline (Km)	Proximity to Hydrogen demand (Km)
4: Most Suitable	> 2400	<10	<5	<5	<3	<10	<20	<5	<10
3: Strongly Suitable	2000–2400	10–20	5–15	5–9	3–10	5–20	20–40	5–20	10–20
2: Suitable	1800–2000	20–60	15–60	9–12	10–40	20–40	40–60	20–40	20–40
1: Moderately Suitable	1500–1800	60–100	60–100	12–16	40–100	40–100	60–72	40–100	40–100

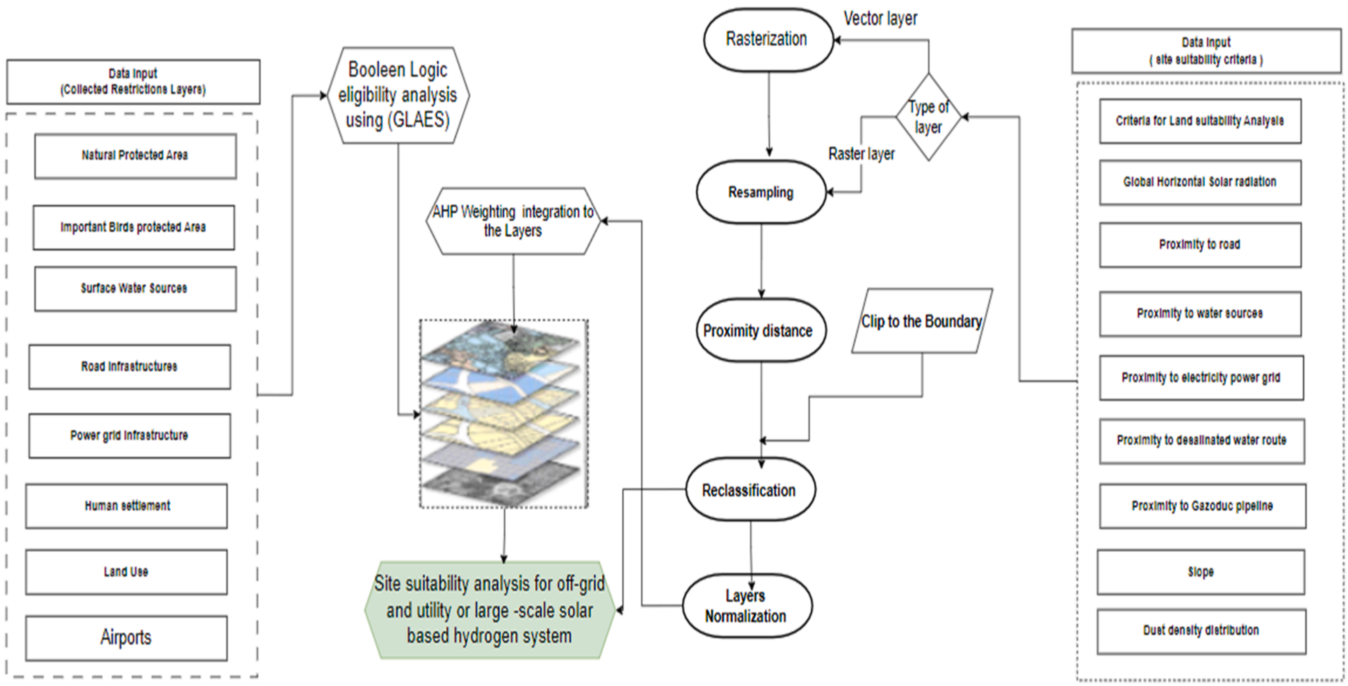

Fig. 3. Land suitability analysis framework workflow.

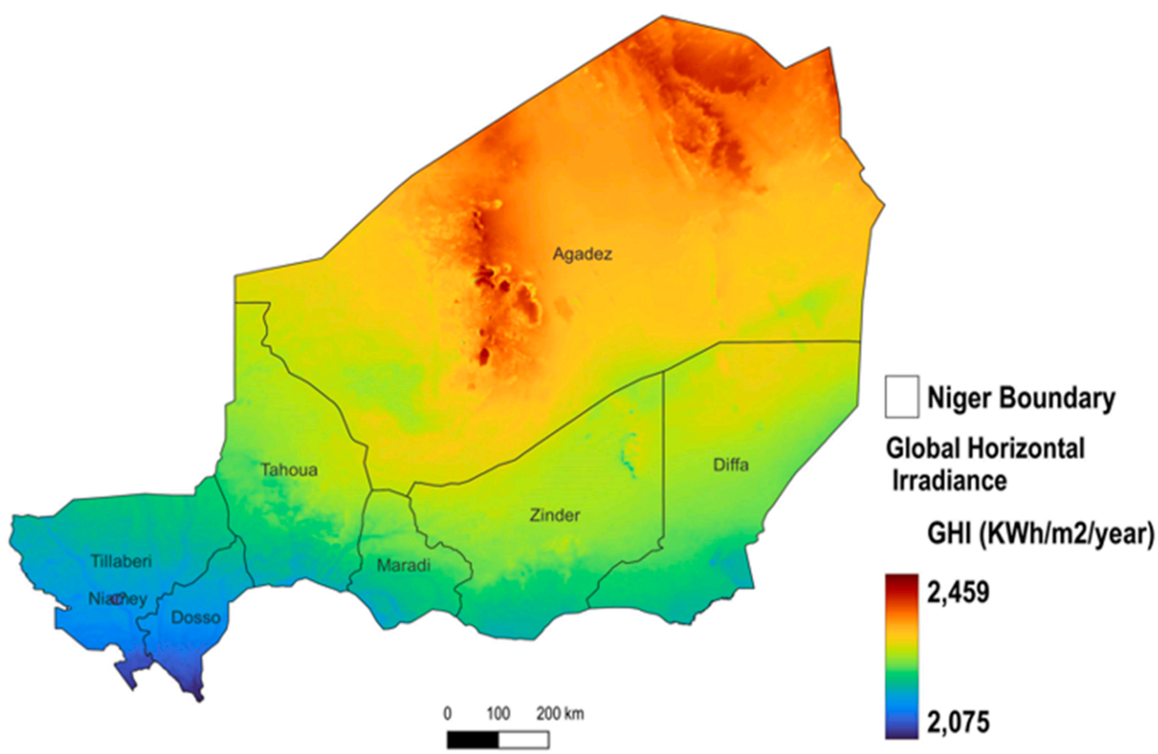

Fig. 4. Proximity distance reclassified map for solar radiation.

country to develop a hydrogen economy and achieve the electrification target. To reduce water transport, the cost of a proximity distance to the water source was calculated for the geospatial site suitability analysis, the Fig. 8 and Fig. 9 represent the reclassified of proximity distance layer respectively fir domestic water source and the hypothetical desalinated water route.

f) **Slope:**

The gradient of the topography can affect the amount of solar radiation received in an area. The slope of the land surface is a crucial factor as far as construction cost is concerned because steep slopes of a surface can reduce the accessibility of trucks and increase building costs. The flat terrain is essential for large-scale PV farms; high-slope areas are not preferable for such projects due to high economic feasibility The slope raster layer is generated from the Digital Elevation Model (DEM) using the raster terrain procession and slop algorithm. Flat areas or mild steep slopes will help to avoid the high

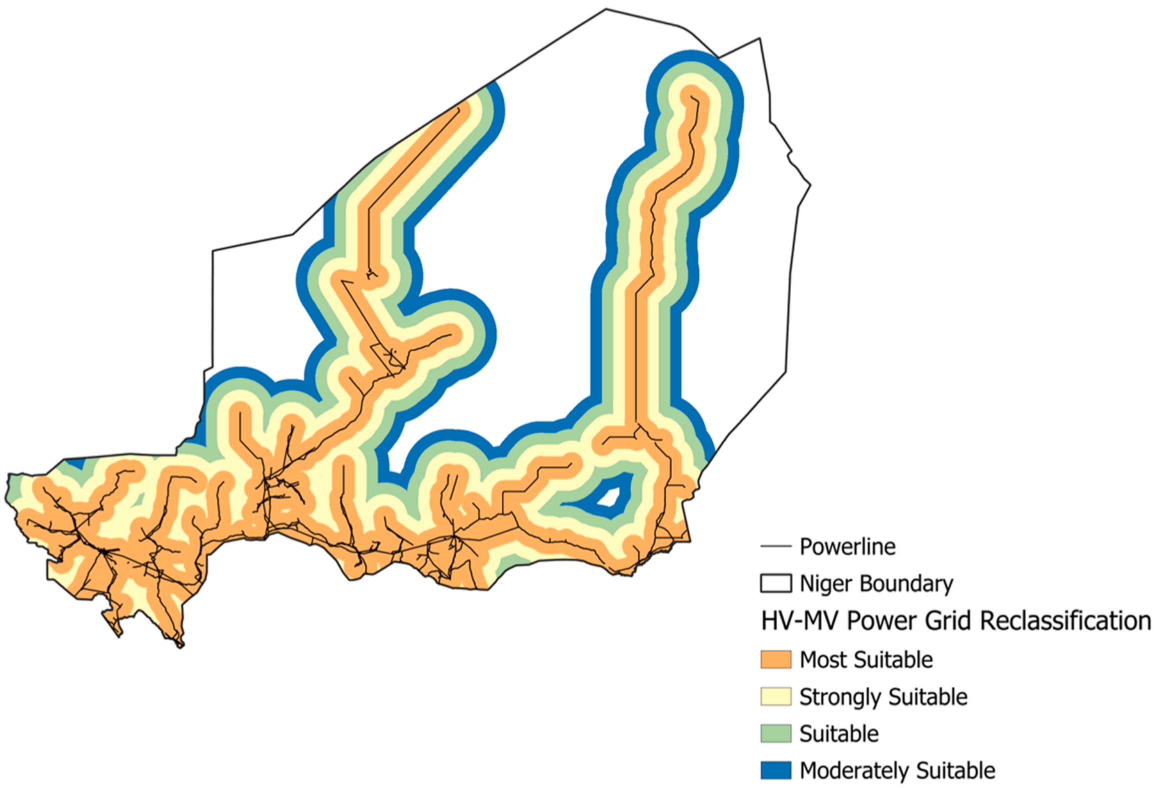

Fig. 5. Proximity distance reclassified map for Power grid.

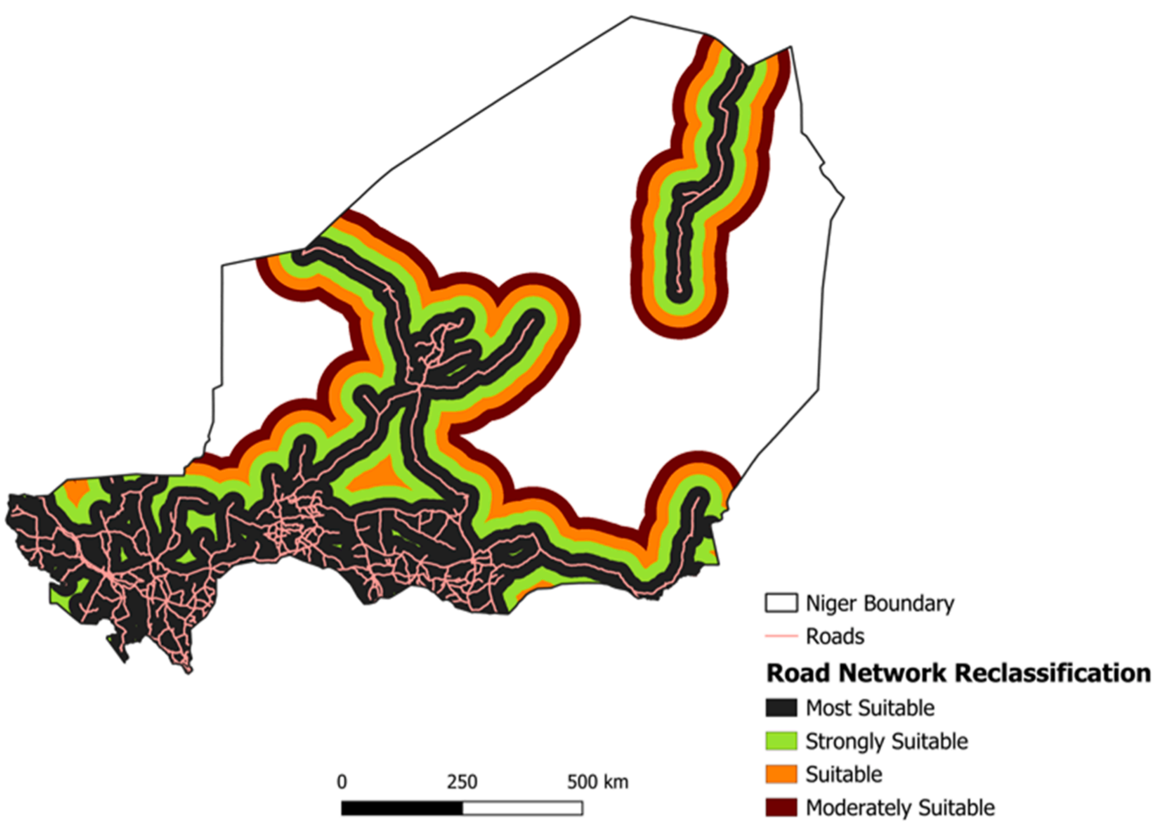

Fig. 6. Proximity distance reclassified map for Road Network.

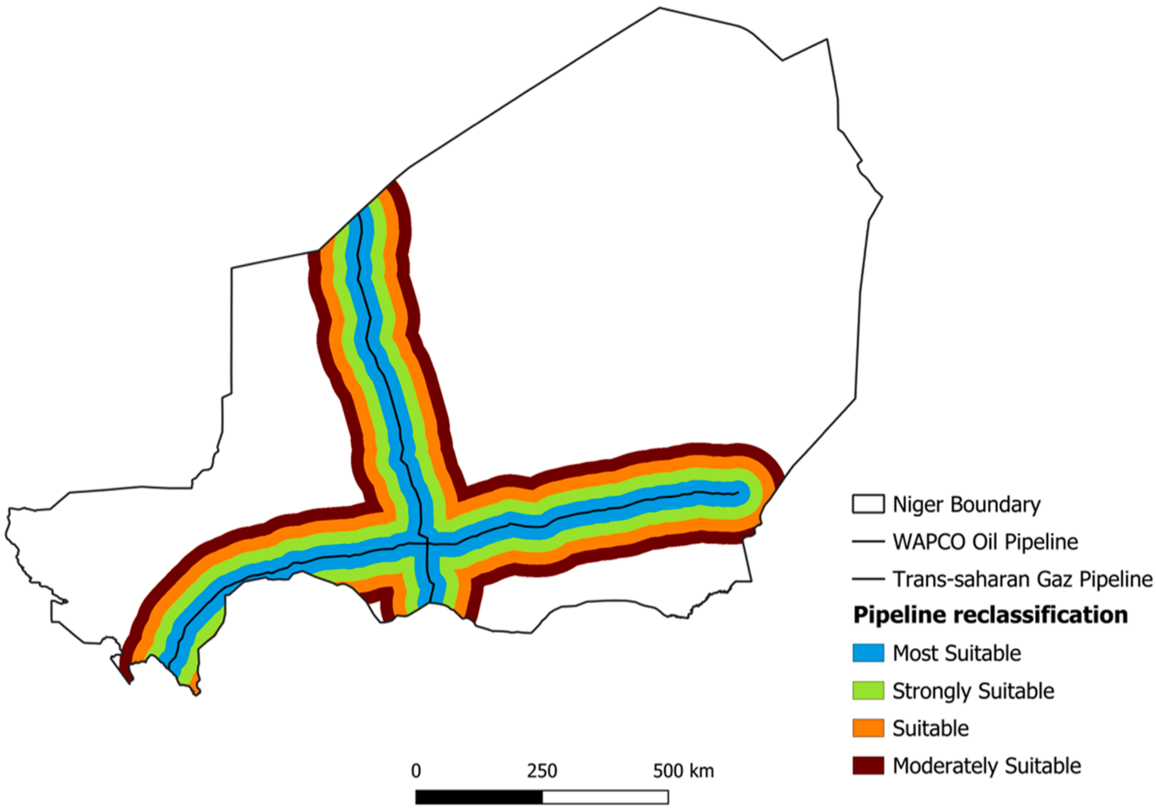

Fig. 7. Proximity distance reclassification for Pipeline infrastructure.

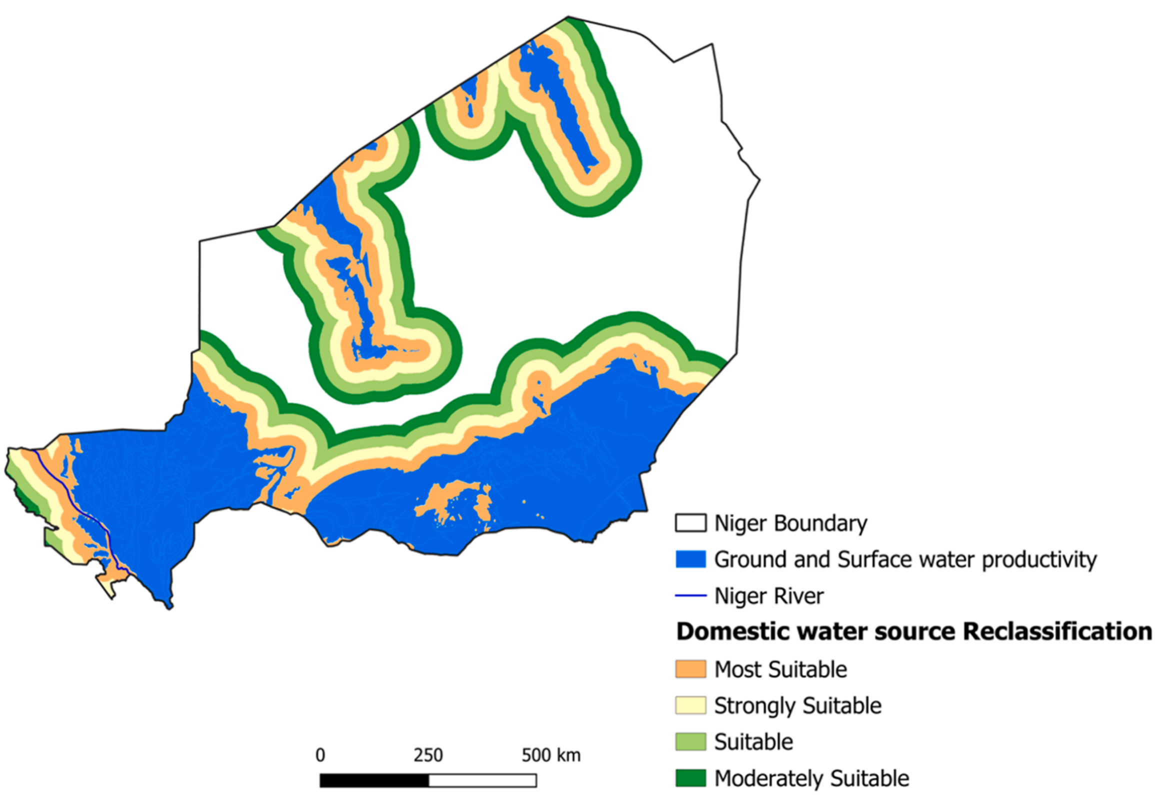

Fig. 8. Proximity distance reclassification for a Water source.

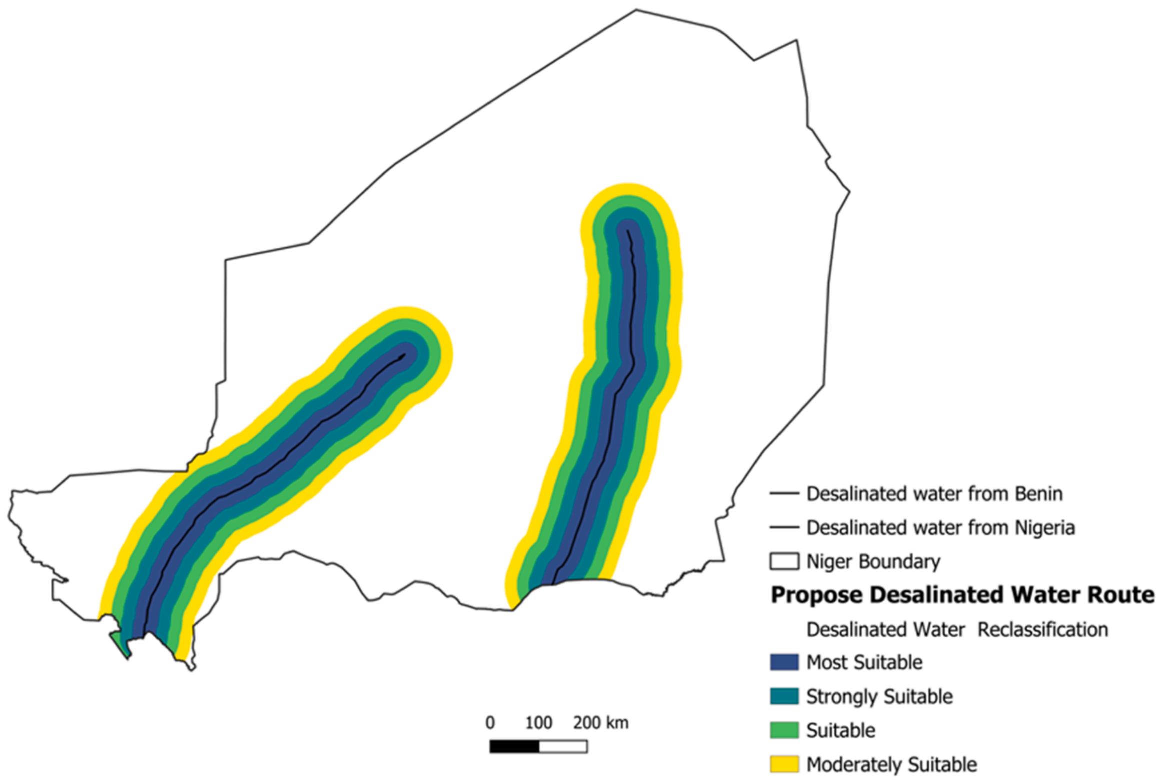

Fig. 9. Proximity distance reclassified map for desalinated water route.

construction cost required in high slope areas. Flat terrain is essential for large-scale PV farms; as such, high slope areas are not preferable for such projects due to low economic feasibility (Zuhair and Garni, 2018). A south-facing slope is an ideal orientation for solar farm sites and must be less than 5 degrees Celsius in this study. Higher slope areas such as valleys and steep lands should be avoided. Using the DEM, the aspects of the survey area have been generated. The obtained slope raster layer varies from 1 to 16.58 degrees Celsius. The

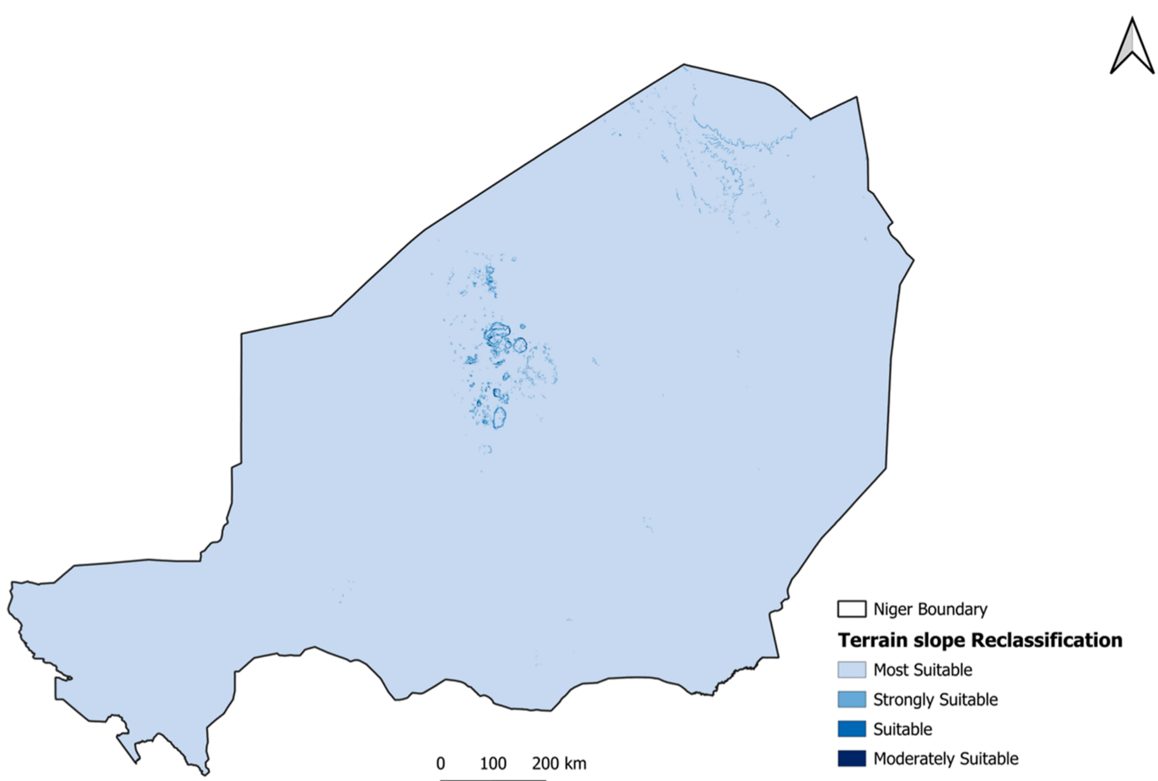

Fig. 10. Proximity distance reclassification for slope.

slope is reclassified into five classes for the suitability analysis. Fig. 10 depict the reclassification of the slope of the terrain. According to a study from IRENA, the landscape must usually not tilt more than 100 % (45 degrees Celsius) for PV systems (IRENA, 2016)). The slope raster layer is extracted from the Digital Elevation Model collected from the CGIAR platform using the QGIS slope calculation tool. Additionally, slope reclassification was performed, and preference was given to lower slopes with higher value scores to increase the economic criteria in the logistic transport. For the suitability analysis, the slope input data concerns only the utility or large-scale system as criteria for site selection.

g) **Dusty area:**

Dust accumulation on solar panels can reduce their efficiency by blocking sunlight and decreasing the amount of energy converted into electricity, which impacts the overall productivity and effectiveness of solar-based green hydrogen production and increases the system's operation and maintenance cost. Mostly in Sahelian dry areas, considering dust accumulation in suitability analysis is worthwhile, although it was not considered in the recent studies (Asare-Addo, 2023; Bayounis and Eldamaty, 2022; Ghasemi et al., 2019). However, dust deposition analysis in Romania shows a density of up to 11.7 g m^{-2} in 2015 (Danu et al. 2018), while in Poland the daily dust density accumulation in the absence of rain can reach up to 40 mg m^{-2} day^{-1} (Jaszczur et al. 2019), in Kraków, a study found a 2.1 % PV efficiency loss after a week with over 300 mg m^{-2} of dust accumulation (Styszko et al. 2019). Moreover, the average monthly surface mass density (Kg/m^2) of the MERRA 2 Model was collected from the NASA GIOVANNI site in raster type (Earth Data, 2023). The dust density in Niger varies considerably, with monthly accumulation ranging from 248 to 721 mg/m^2 and weekly averages between 62 and 180 mg/m^2 (Earth Data, 2023). This high level of dust, primarily originating from the region's sandy soils and desert environments, has a significant impact on photovoltaic efficiency and necessitates increased maintenance costs for power plants. Recognizing the importance of accounting for dust in site suitability assessments, this study involves reprojecting and resampling raster-based dust density data to identify optimal locations for utility-scale and large-scale solar hydrogen production facilities that minimize exposure to dust. The dust density raster layer is reclassified for the overlay analysis of the suitable site in the Fig. 11.

h) **Population density:**

Population density is a relevant factor for suitability analysis as it describes the energy demand of the location. Higher population density means significant energy needs, mainly in urban and peri-urban areas. The lower density is sparse, and the lower electricity demand area is remote from the grid infrastructures. Raster-based population density data are collected from the WorldPop (Tatem , 2017). In order to evaluate the site suitability according to the size of the density, population data is clustered using Density-Based Spatial Clustering of Applications with Noise (DBSCAN) by adapting the method developed using Python, the code is available on GitHub (Khavari et al., 2021; Korkovelos et al., 2019). Depending on the type of technology the insight from the survey, the analysis is conducted in such a way that the off-grid system closer to the settlement is preferable in contrast locations far away from the settlements is more suitable. The optimal location for a green hydrogen production suitability analysis depends on the technology and the insights from the survey. Systems closer to settlements may be preferred for off-grid system, while remote locations from the settlements might be more suitable for large-scale systems.

3.2.3. *Criteria weighting using the Analytic Hierarchy Process (AHP) method*

The AHP method is based on mathematics and psychology, introduced in 1980 by Saaty (Saaty, 1980). In the AHP procedure, the weight of each criterion is evaluated using the pair-wise comparison approach. A weight is a value assigned to an evaluation criterion that indicates its importance relative to the other criteria under consideration (Malczewski, 2015). AHP uses hierarchical structures to illustrate a problem and help make judgements based on expert advice to deduce

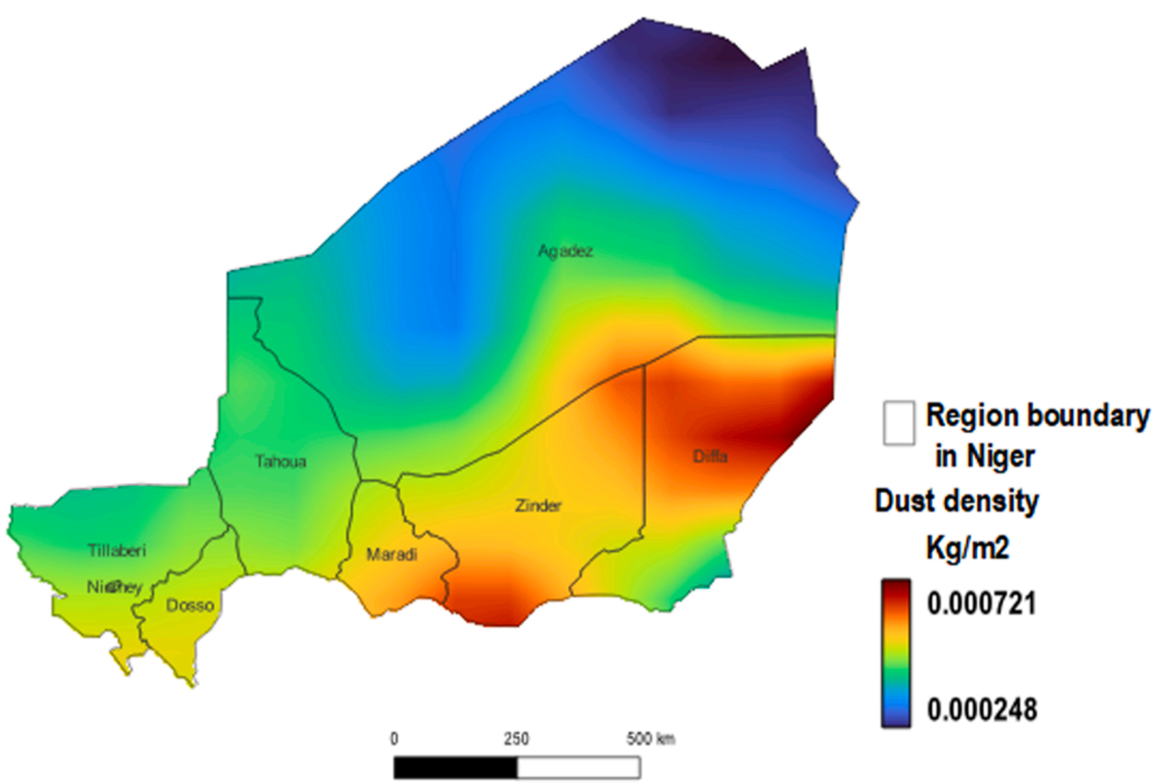

Fig. 11. Proximity distance reclassified map for dusty area.

priority scale. Qualitative and quantitative inputs can be combined in AHP to resolve complex problems in energy planning. The robustness and flexibility of the AHP method in the MCDA prove the broader use in renewable energy studies, according to Zarghanmi (Zarghami and Szidarovszky, 2011). The method relies on a pair-wise comparison of evaluation criteria using a fundamental nine-point scale measurement to express the intensity of importance (Zadeh, 1965) with a reciprocal ratio. The pair-wise criteria are based on the importance of the factor regarding the type of system. Accordingly, a survey and expert interviews are carried out for criteria weighting based on whether the system is off-grid, grid-connected or large-scale solar–based hydrogen production.[2]

Assuming there are n criteria, a pair-wise comparison matrix A (n x n) is generated based on the judgments from specialists, where the element presents the importance of the ith criterion to the jth criterion. Accordingly, the relative importance of the jth criterion to the ith is reciprocal of $a_{ij,}$ i.e. $a_{ji}= 1/\ a_{ij}$. A scale of 1–9 is used to measure the relative importance, where 1 means equal importance, and 9 is extremely higher importance of one criterion than another (Sun et al., 2020).

Table 4 represents the judgment scale of the pair-wise comparison matrix generation, where the value on the left of the matrix is the reciprocal value of the pair-wise criteria comparison. The matrix below provides further insight into carrying out AHP. The consistency index (CI) is determined and compare to the consistency ratio (CR) to overcome the arbitrary and inconsistent decision, the average value of the inconsistency index RI is shown in the Table 5.

$$A = \begin{bmatrix} 1 & a & b \\ 1/a & 1 & c \\ 1/b & 1/c & 1 \end{bmatrix} \quad (2)$$

The consistency index is computed using the formula (3)

$$CI = \frac{\lambda_{\max} - n}{n-1} \quad (3)$$

Where $\lambda_{\max}$ is the *maximum* eigenvalue of the comparison matrix A.

the consistency ratio (CR) is defined as the ratio of the consistency index CI to the average consistency index RI, given by Thomas Saaty (Saaty, 1980) in the Eq. (4).

$$CR = \frac{CI}{RI} \quad (4)$$

Thus, the value of CR reflects the correctness of the procedure when the consistency ratio is lesser or equal to 0.1 ($CR \leq 0.1$). Therefore, the level of consistency is acceptable; otherwise, a major adjustment of the pair-wise comparison values is required (Sun et al., 2020). Further, the random index (RI) represents the average deviation from randomly generated matrices of different sizes (Asare-Addo, 2022). The pair-wise comparison matrix processing is performed using the Python-based AHP algorithm developed by Philip, available on GitHub (Griffith, 2021). As a result, the program processes the pair-wise comparison matrix to calculate the criteria weighting and the consistency ratio to be compared with the standard value.

4. Results and discussions

The present study developed a comprehensive framework to assess land suitability for evaluating solar hydrogen sites. The analysis is being carried out for off-grid, grid-connected (utility), and large-scale systems.

Table 5
Value for average inconsistency index RI.

n	2	3	4	5	6	7	8	9
RI	0.00	0.58	0.90	1.12	1.24	1.32	1.41	1.45

An extensive geographical potential for site suitability analysis is presented. The study summarises the eligible land for implementing a solar hydrogen system and ranks sites regarding their suitability according to the aforementioned criterion, further sensitivity to the criterion weighting is conducted using equally weighting approach. The overarching objective of this proposed model is to offer decision support for energy policy planning, specifically focusing on developing a hydrogen economy in Sub-Saharan Africa (SSA).

4.1. *Land eligibility analyses*

The results on land eligibility using GLAES highlights that protected natural areas cover 16 % of Niger territory, located mainly in the northeast (Zinder, Diffa and Agadez). Although the population are sparsely distributed around the country, human settlement represents the second most significant constraint of site eligibility for hydrogen production. As depicted in Fig. 12, The result of the land eligibility analysis shows that the excluded constraints represent 32.11 % of the country's land mass. In contrast, 67.89 % (889,834 square kilometer) is available for installing solar hydrogen systems. This area represents twice the size of Germany, making Niger well-positioned to explore the implementation of solar based hydrogen economy. The distribution of site eligibility through the country is presented in Fig. 13. Therefore, due to its vastness and low density of the population, the Agadez region represents 78 % (548,217 square kilometer) of eligible land. Conversely, only 5 % of the Niamey region is available for solar PV power plant installation for hydrogen production, which faces limitations due to high population density, settlement, and land use for agriculture.

4.2. *Land suitability for off-grid solar-based hydrogen production*

The off-grid hydrogen production site suitability criteria include global horizontal irradiance (GHI), road proximity, water availability, dust levels, proximity to settlements, and distance from the power grid. The criteria were divided into two main factors: technical and economic. Technical criteria include solar radiation potential, hydrogen demand, and slope, while economic criteria comprise road proximity, water access, and power grid distance. The analysis of the pair-wise matrix (table A.1 Appendix A) indicated that hydrogen demand corresponding to the population density accounts for 0.4 of the total weight, closely followed by water demand with a weight of 0.26, as shown in Table 6.

In contrast, the proximity to the power grid received the lowest weight, representing 0.037, since areas closer to the grid are avoided for off-grid installation. Furthermore, proximity to roads is crucial economically, leading to a higher weight allocation in areas closer to main roads to reduce installation and maintenance costs. Global horizontal irradiance holds less weight than water availability and hydrogen demand due to its typically high average yield, making those areas less challenging. The calculated consistency ratio is 0.033, confirming the consistency of the pair-wise matrix according to the Saaty AHP method (Saaty, 1980).

As illustrating in the Fig. 14, the land suitability for potential off-grid solar-based hydrogen production sites is reclassified from moderately

Table 4
Importance Scale.

1/9 **Extremely**	1/7 **Very strongly**	1/5 **Strongly**	1/3 **Moderately**	1 **Equally important**	3 **Moderately**	5 **Strongly**	7 **Very strongly**	9 **extremely**
Less important→More import								

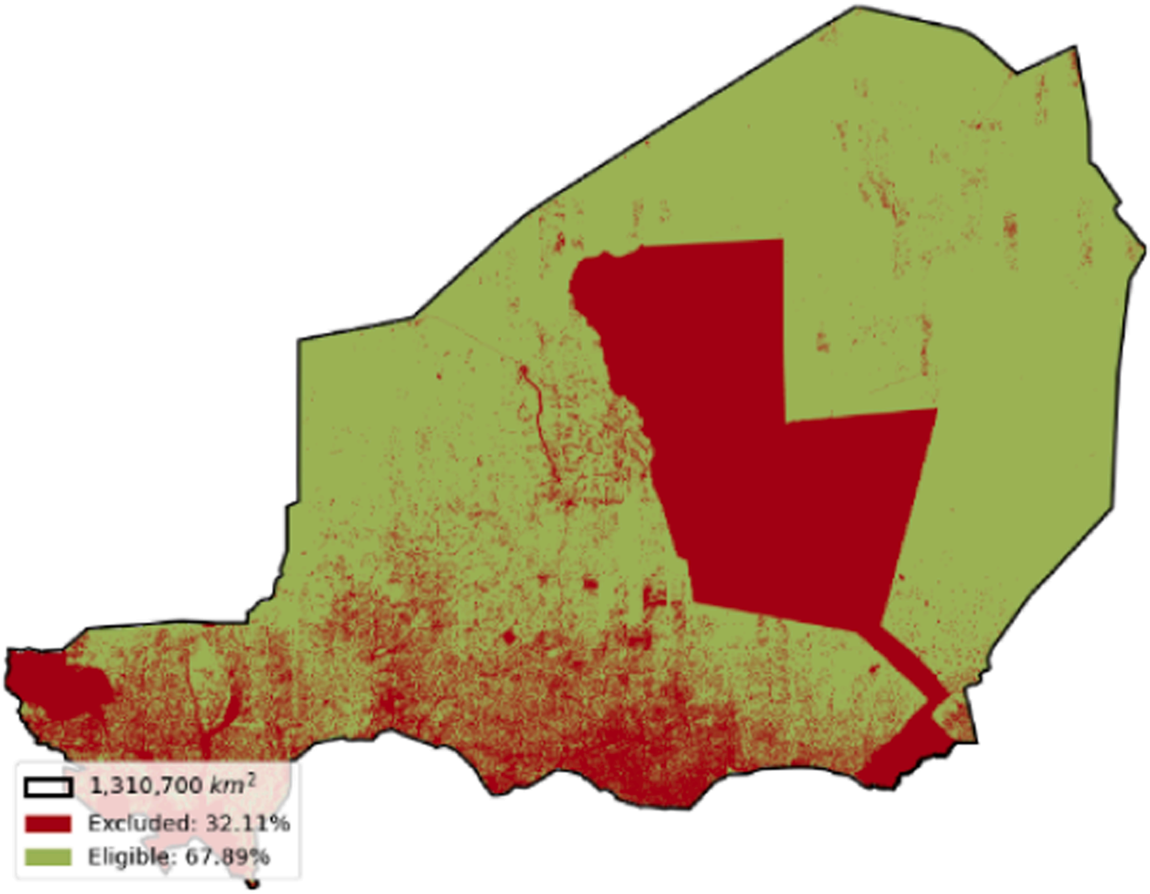

Fig. 12. Land eligibility for solar hydrogen potential evaluation.

Fig. 13. Solar hydrogen production site eligibility analysis per region in Niger.

suitable to highly suitable based on the pixel value of the overlay output layer similarly in Tafula et al., (2023a).

The analysis results indicate that 32.36 % (421,842 square kilometer) of the area is suitable (orange) for implementing off-grid solar PV systems for hydrogen production. Additionally, 2.19 % (28,621 square kilometer) is moderately suitable (light orange), while 26.1 % (340,184 square kilometer) show strong suitability (light green). Conversely, the most suitable areas (dark green) encompass 7.05 % (91,920 square kilometer) and are primarily located in the eastern and northern regions of Niger. These areas feature proximity to road networks, abundant solar

Table 6
Summary of the criteria and sub-criteria weight for site suitability for off-grid solar-based hydrogen production.

Main Criteria	Weight	sub-criteria	Wi	Weight (%)
Technical criteria	0.582	GHI	0.213	12.4
		Slope	0.087	5.1
		Hydrogen demand	0.699	40.7
Economic Criteria	0.418	Proximity to water	0.624	26.1
		Proximity to Power grid	0.088	3.7
		Proximity to roads	0.289	12.1

radiation, ample water resources, and high hydrogen demand while distant from the power grid. In contrast, the moderately suitable zones (orange) are predominantly situated in the central and northern regions, characterised by lower population density, limited road infrastructure, and scarcity of water resources.

4.3. Land suitability for Grid-connected solar-based hydrogen production

The criteria selection relevant to grid-connected systems for solar-based hydrogen production suitability analysis was undertaken through interviewed surveys followed by criteria weighting as in the pair-wise comparison matrix table A.2 in appendix A. Among the selected factors, economic criteria encompassed proximity to the power grid, roads, and water sources, while technical criteria involved factors such as global horizontal irradiance (GHI), settlement availability, dust, and terrain slope. The pair-wise comparison matrix calculation highlights the increase in the economic criteria weight of around 0.631, as shown in *Table 7*. Subsequently, proximity to the power grid held the highest weight (0.228) among economic criteria, followed by proximity to water sources and roads (0.193 and 0.192, respectively), indicating a potential for reducing hydrogen production cost and enhanced competitiveness with fossil fuel. The technical factors weight 0.387 of the overall criteria, with a distinct emphasis placed on solar radiation (0.218). A calculated consistency ratio of 0.071 aligned with the Saaty AHP threshold (Saaty, 1980).

The overlay analysis of the different criteria layers depicted in the map of Fig. 15 presents the reclassified site suitability demonstrates that 7.3 % (95,658 square kilometer) of the eligible area has sites deemed suitable. In contrast, only some areas are allocated as moderately suitable. The strongly suitable sites (light green) for grid-connected solar-based hydrogen production systems correspond to 44.69 % (585,414 square kilometer). They are attributing to the more significant solar radiation potential in the northern regions and the presence of groundwater resources in the west-northern areas. Additionally, the central region benefits from proximity to main roads and ample solar radiation, contributing to this higher suitability. Most site suitability (dark green) covers 15.79 % (206,945 square kilometer) of Niger's territory eligible land. It can be noticed that the highly suitable areas exhibit more substantial economic factors, implying shorter distances to water resources, power grids, roads, and higher solar radiation.

4.4. Land Suitability for Large-scale solar-based hydrogen production using domestic water sources

In the land suitability assessment for large-scale solar-based

Table 7
Summary of the criteria and sub-criteria weight for site suitability for Grid-connected solar-based hydrogen production.

Main Criteria	Weight	sub-criteria	Wi	Weight (%)
Technical criteria	0.387	GHI	0.563	21.8
		Dust	0.147	5.7
		Population	0.206	8
		Slope of the terrain	0.082	3.2
Economic Criteria	0.613	Proximity to water	0.313	19.2
		Proximity to Power grid	0. 371	22.8
		Proximity to roads	0.314	19.3

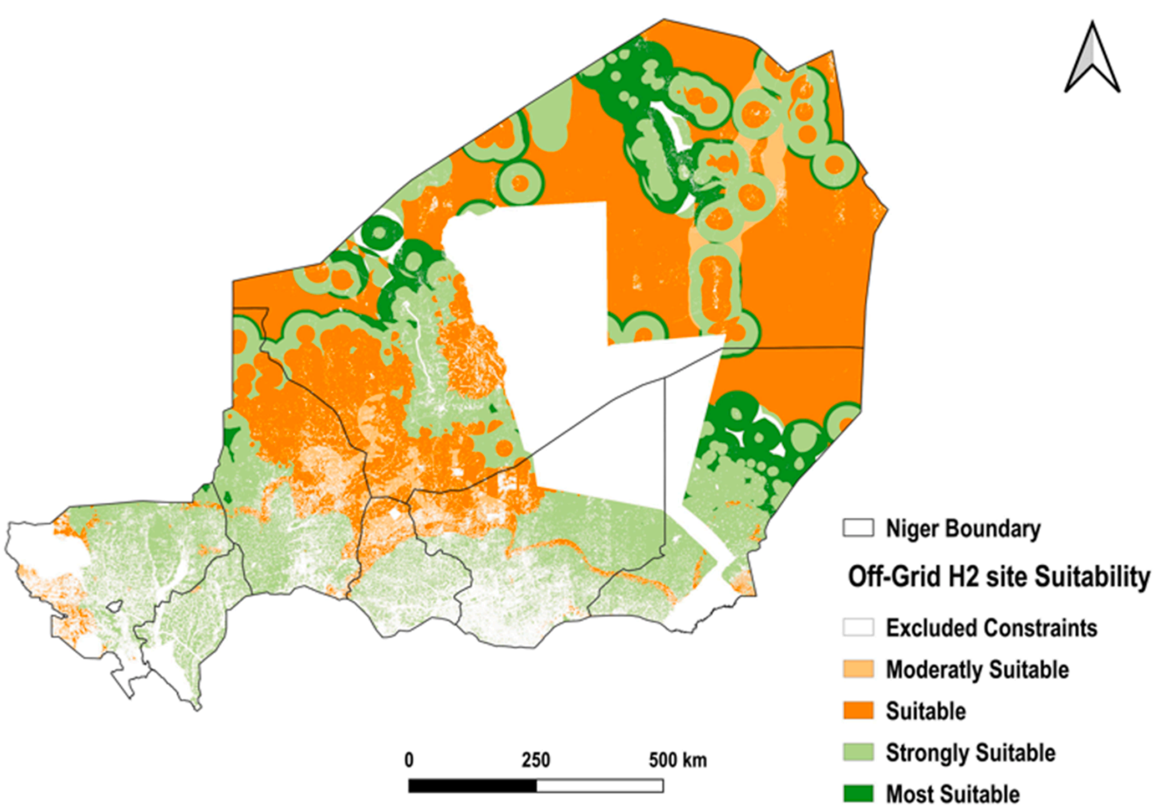

Fig. 14. Off-grid solar-based hydrogen production potential site suitability.

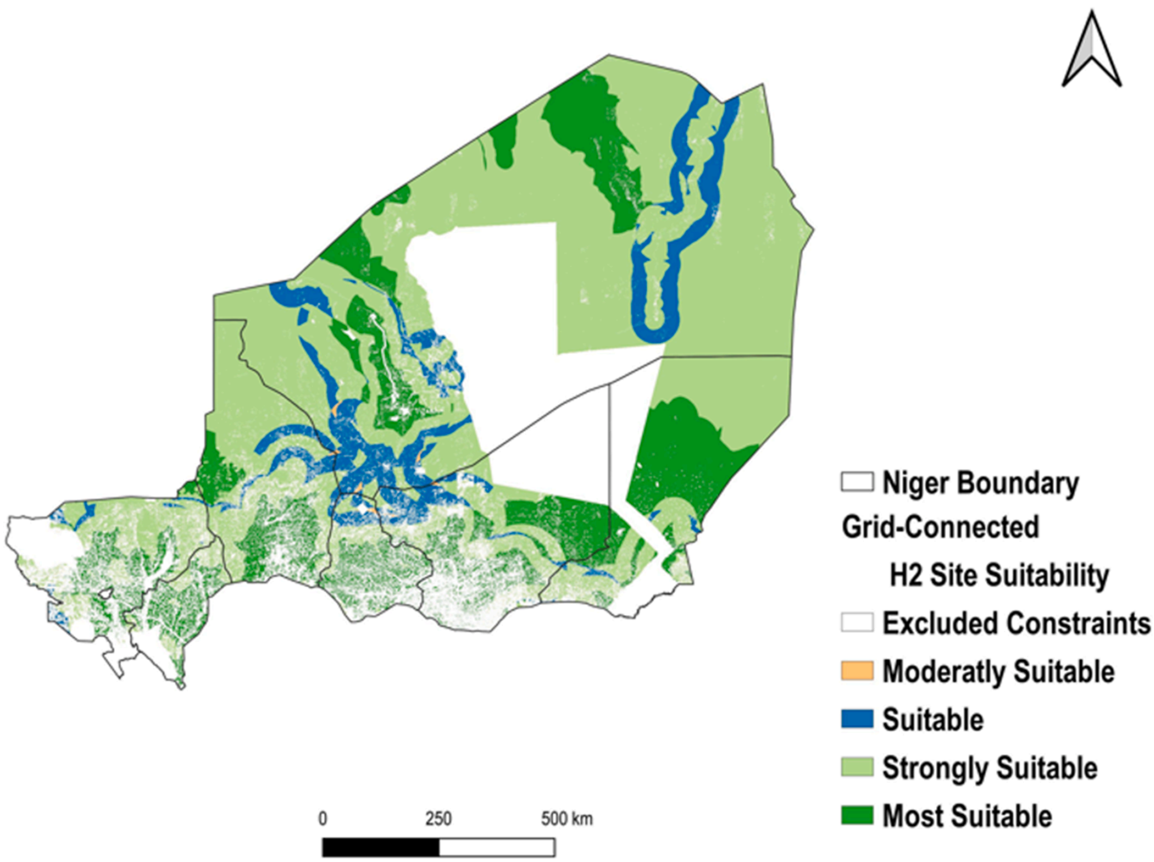

Fig. 15. Grid-connected solar-based hydrogen production potential site suitability.

hydrogen production for export, domestic water sources, proximity to roads, water sources, human settlements, and pipelines are categorised as economic criteria. At the same time, dust levels, slope, and solar irradiation are technical criteria. Criteria weighting is applied to economic and technical criteria, accounting for 0.544 and 0.456, respectively. Additionally, the viability of hydrogen technology is hindered by the challenge of transporting it for export. The proximity to water importation pipelines is crucial for identifying suitable sites, with a weight of 0.278, followed by solar potential, with a weight of 0.19. As shown in Fig. 7, the Trans-Saharan gas pipeline runs through Niger, connecting Nigeria to Algeria. This pipeline could facilitate the export of hydrogen to Europe.

Furthermore, recognising the adverse effects of dusty environments on solar system maintenance and fuel cell lifespan, dusty area weight 0.186 using the pair-wise comparison. The lower weight for proximity to roads is not due to reduced importance (Table 8). Instead, it reflects the sparse road network, which could be addressed by constructing new roads to improve access to potential sites for solar hydrogen production. The obtained consistency ratio of 0.07 aligned with the standard threshold (Saaty, 1980).

Table 8
Summary of the criteria and sub-criteria weight for site suitability for large-scale solar-based hydrogen production using domestic water source.

Main Criteria	Weight	sub-criteria	Wi	Weight (%)
Technical criteria	0.456	GHI	0.41	19
		Dust	0.407	18.6
		Population	0.111	5.1
		Slope of the terrain	0.063	2.9
Economic Criteria	0.544	Proximity to water	0.511	27.8
		Proximity to roads	0.161	8.8
		Proximity to Pipeline	0.327	17.8

The map of Fig. 16 depicts the land suitability assessment for a large-scale solar hydrogen system, focusing on utilising domestic water sources (river + groundwater). The overlay analysis of the layers according to technical and economic criteria weights show that 7.62 % (99 971 square kilometer) is suitable (pink colour), 20.05 % (262 794 square kilometer) has strong suitability (light green), while 40 % (526 939 square kilometer) of the eligible land has been found most suitable (dark green) primarily situated in the northern and west-northern regions. This suitability can be attributed to higher solar radiation, access to groundwater resources, and distance from dusty areas.

4.5. *Land suitability for large-scale solar-based hydrogen production using desalinated water sources*

The potential site suitability for hydrogen production from large-scale solar power plants was analysed using desalinated water. The water source was considered to be from two pathways Nigeria and Benin. Therefore, a desalinated water route was suggested based on the proximity to the trans -Saharan gas pipeline and the Niger-Benin oleoduc (see Fig. 7). This could be explained by the fact that a feasibility study had already been carried out before choosing the proposed pathway.

The criteria weighting using an AHP-based pair-wise comparison matrix shows that GHI is the most significant technical criterion, which includes terrain slope, dusty areas, and population density, accounting for 26 % of the total technical criteria weight. Regarding the economic criteria (see Table 9), proximity to the water transport sources has the highest weight, followed by the road and pipeline future infrastructure route for hydrogen export. The pair-wise comparison of the input criteria using desalinated water for large-scale solar-based hydrogen production is summarised in table B.2 of Appendix B.

The results of the Fig. 17 demonstrate that 54.14 % (709,539 square

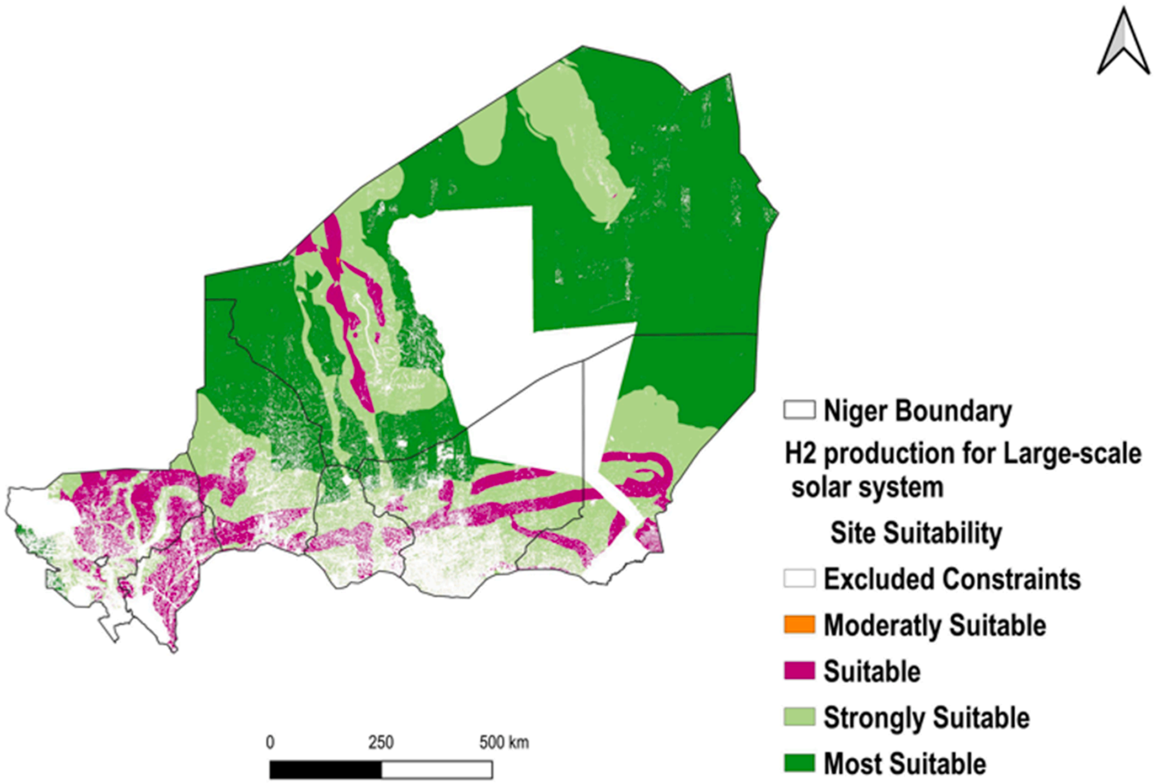

Fig. 16. Land Suitability for Large-scale solar-based hydrogen production from domestic water sources.

Table 9
Summary of the criteria and sub-criteria weight for site suitability for large-scale solar-based hydrogen production using desalinated water.

main Criteria	Weight	sub-criteria	relative weight Wi	Weight (%)
Technical criteria	0.396	GHI	0.65	26
		Dust	0.156	6.2
		Population	0.106	4.2
		slope of the terrain	0.08	3.2
Economic Criteria	0.604	Proximity to water	0.58	35.5
		Proximity to roads	0.314	9
		Proximity to Pipeline	0.263	15.9

kilometer) is strongly suitable (green colour) for exporting large-scale solar-based hydrogen production. While only 2.05 % (34,822 square kilometer) is deemed mostly suitable, and 11.05 % (144,858 square kilometer) represents the suitable area (blue colour). It can be noticed that no area is allocated as moderately suitable (orange colour). Furthermore, the most suitable areas align closely with the proposed water transport infrastructure, whereas the strongly suitable zones are dispersed across the West-Northern and North-Eastern regions. Conversely, the West-Southern part exhibits lower suitability due to the presence of settlements, potentially impeding the installation of large-scale solar hydrogen systems meant for export. It has been found that the literature has yet to explore desalinated water importation scenarios in hinterland countries.

4.6. Sensitivity analysis

In the calibration of the framework for site suitability analysis for green hydrogen production a sensitivity analysis is applied to find out the implication on the siting reclassification. Therefore, the equally criterion weight approach is conducted for the overlay analysis for the large-scale green hydrogen system for export using domestic water sources (surface & Ground water) called large-scale and using desalinated water (large-scale), for grid-connected and off-grid systems. The Fig. 18 represents the equally suitability analysis criterion weight, as we can observed the large part of the land available is deemed suitable for solar-based green hydrogen production representing more than 47 % (624 421 square-kilometer) of the land-masse, thus could be explained by the suitable orographic of the area and the high sun irradiance well distributed throughout the country. In constrast, the most suitable site represents less than 1 % of the of the eligible area, while 3–13 % of the area is strongly suitable according the type of the plant, this is mainly demonstrated by the low distribution of the infrastructure over the territory mostly in the north-eastern part of the country. The criterion selection survey and the insight from the expert are summarized in the appendix C of the figures.

5. Conclusion

The sustainable development of renewable energy sources, such as green hydrogen, is a multifaceted task that involves technical, economic, and environmental criteria. The participation of various stakeholders adds complexity to the task. When the involvement of multiple stakeholders is not adequately addressed, conflicts of interest may arise, compromising the project's sustainability goals.

In this study, Geographical Information System and multi-criteria decision-making are combined to develop a framework for solar-based hydrogen site suitability analysis in Niger by effectively addressing the environmental, socio-economic and technical implications. To investigate site suitability, first, a land eligibility analysis was carried out using the GLAES tool, followed by a land suitability assessment for hydrogen potential evaluation, which considered technical and economic criteria. The results of the solar-powered hydrogen potential site suitability show that:

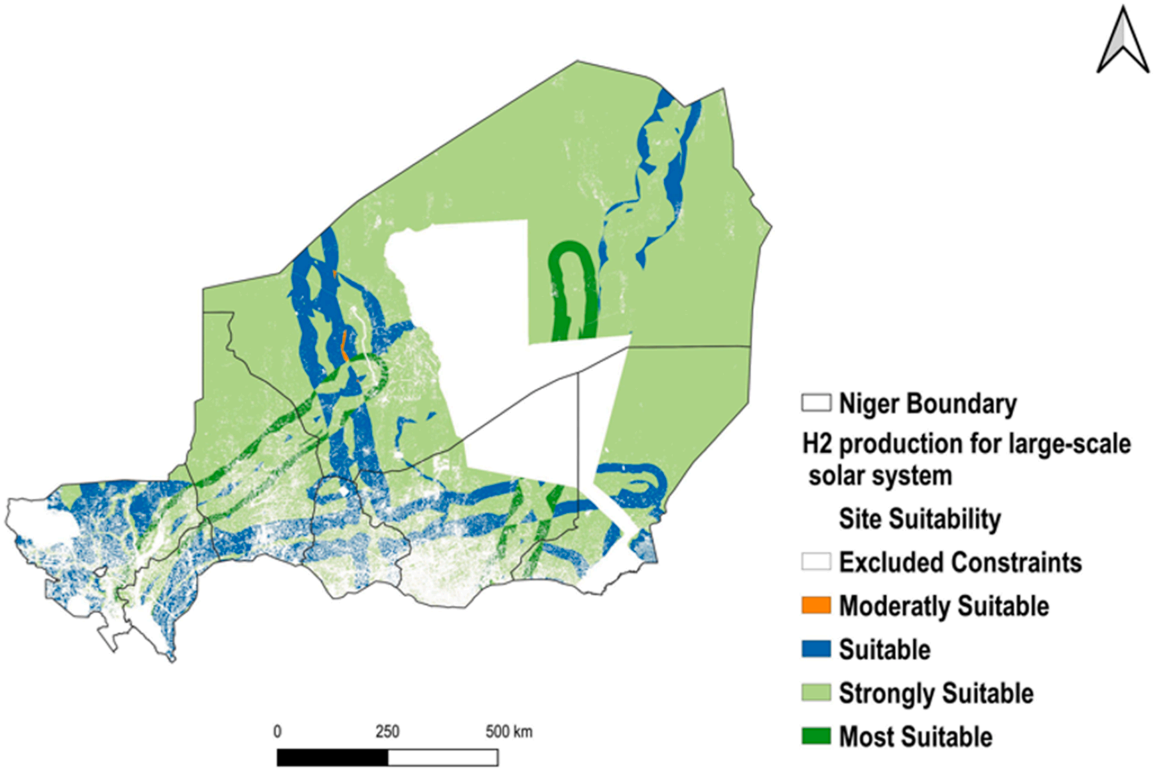

Fig. 17. Land Suitability for Large-scale solar-based hydrogen production from desalinated water sources.

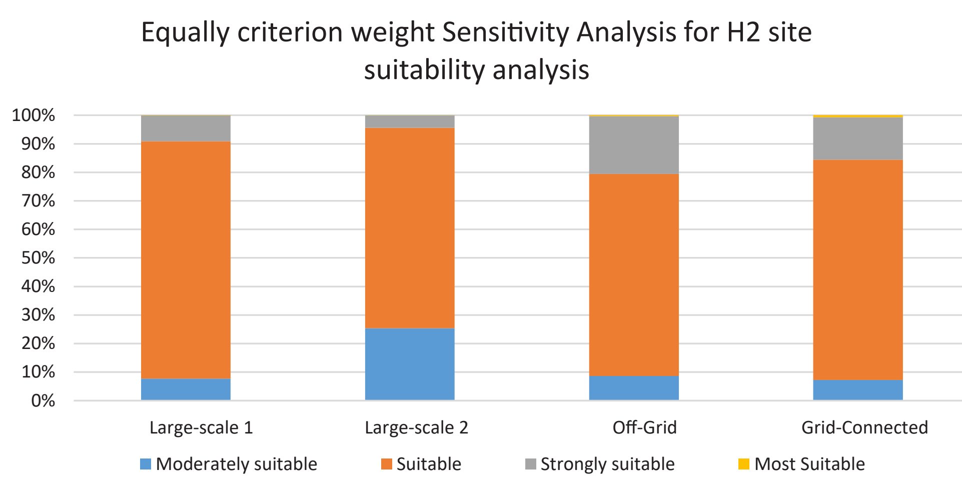

Fig. 18. Equally criterion weight sensitivity analysis fir green hydrogen production site suitability assessment.

- Land eligibility analysis shows that 67.89 % (889,834 square kilometer) is deemed eligible while 32.11 % (420,738 square kilometer) represent excluded constraints,
- The land suitability analysis for off-grid system shows that 32.36 % (421,841 square kilometer) has low suitability, 2.19 % (28 621 square kilometer) is deemed moderately suitable, 26.1 % (340,183 square kilometer) of the eligible area has strongly site suitability while 7.05 % (91,920 square kilometer),
- land suitability analysis for Grid-connected demonstrates that 7.3 % (95,658 square kilometer), 44.69 % (585,414 square kilometer) are strongly suitable; in contrast, the most suitable site represents 15.79 % (206,945 square kilometer),
- The analysis for solar-based hydrogen production using domestic water sources shows that 7.62 % (99,971 square kilometer) is suitable, 20.05 % (262,794 square kilometer) has strong suitability, while 40 % (526,939 square kilometer) of the eligible land has been found most suitable primarily situated in the northern and west-northern regions,
- Large-scale hydrogen production system suitability analysis using desalinated water reveals that 54.14 % (709,539 square kilometer) of the eligible land is strongly suitable for large-scale solar-based hydrogen production for export.

Moreover, by addressing overlooked constraints to geographical factors, our study provides a more comprehensive view of spatial techno-economic cost constraints. As a limitation, the techno-economic potential to determine the impact of the geographical criteria on the levelized cost of hydrogen is out of the scope of this study. Although these insights serve as a roadmap for policymakers, and energy planners, aligning strategies with ECOWAS hydrogen targets and fostering collaborations for sustainable energy transitions, therefore it is worth investigating in further research the techno-economic over the green hydrogen supply-chain on the cost of hydrogen using geospatial approach

CRediT authorship contribution statement

Ramchandra Bhandari: Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Moussa Mounkaila Saley:** Writing – review & editing, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rabani Adamou:** Writing – original draft, Validation, Supervision, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Bachirou Djibo Boubé:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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Appendix A

Table A.1
AHP pair-wise criteria comparison weighting for off-grid solar-based hydrogen potential site suitability

Criteria	Grid	Road	GHI	Water	Pop	Dust	Weights(%)
Grid	1	0.25	0.33	0.25	0.11	0.5	**3.7**
Road	4	1	0.5	0.5	0.33	3	**12.1**
GHI	3	2	1	0.33	0.25	2	**12.4**
Water	5	2	3	1	0.5	7	**26.1**
Pop	9	3	4	2	1	8	**40.7**
Dust	2	0.33	0.5	0.14	0.125	1	**5.1**

Table A.2
AHP pair-wise criteria comparison weighting for Grid-connected solar-based hydrogen potential site suitability

Criteria	Grid	Road	GHI	Water	Pop	Dust	Slope	Weights (%)
Grid	1	2	0.5	0.5	5	5	7	**22.8**
Road	0.5	1	0.5	2	3	4	6	**19.3**
GHI	2	2	1	1	2	2	4	**21.8**
Water	2	0.5	1	1	2	3	5	**19.2**
Pop	0.25	0.33	0.5	0.5	1	2	3	**8**
Dust	0.25	0.25	0.5	0.33	0.25	1	2	**5.7**
Slope	0.14	0.16	0.25	0.2	0.33	0.5	1	**3.2**

Appendix B

Table B.1
AHP pair-wise criteria comparison weighting for large-scale solar-based hydrogen potential site suitability using domestic water sources

Criteria	Road	GHI	Water	Pop	Dust	Slope	Pipeline	Weight (%)
Road	1	0.33	0.33	3	0.5	0.25	0.33	**8.8**
GHI	3	1	0.5	0.2	0.33	6	2	**19**
Water	3	2	1	5	2	7	2	**27.8**
Pop	0.33	0.2	0.2	1	0.5	2	0.33	**5.1**
Dust	2	3	0.5	2	1	5	0.5	**18.6**
Slope	0.25	0.16	0.14	0.5	0.2	1	0.2	**2.9**
Pipeline	3	0.5	0.5	3	2	5	1	**17.8**

Table B.2
AHP pair-wise criteria comparison weighting for large-scale based hydrogen potential site suitability using desalinated water

criteria	Road	GHI	Water	Pop	Dust	Slope	Pipeline	Weight (%)
Road	1	0.33	0.25	3	2	4	1/3	**9.4**
GHI	3	1	0.5	6	4	7	3	**26**
Water	4	2	1	7	5	9	3	**35.5**
Pop	0.33	0.16	0.14	1	1/2	2	0.33	**4.2**
Dust	0.5	0.25	0.2	2	1	3	0.25	**6.2**
Slope	0.25	0.14	0.11	1/2	1/3	1	0.2	**2.8**
Pipeline	3	0.33	0.33	3	4	5	1	**15.9**

Appendix C

Criteria selection interview and expert insight for solar based green hydrogen site suitability analysis

⟨⟨*Selection a criterion for suitability analysis doesn't mean that the criterion will have a posed impact on the suitable site but although could also impact negatively the efficiency of the site. For instance, the proximity to the power grid to be lower is preferable for grid-connected while the latter is should not be considered for off-grid system to power a remote location*⟩⟩

⟨⟨ *The dust density deposit impact on the site suitability analysis depends on the type of system. The large the power plant (large-scale and grid-connected) is more would the impact of dust on the efficiency on the power plant and incurred cost of maintenance (cleaning); smaller is the plant (off-grid) easer would be the dust cleaning consequently lower maintenance cost.* ⟩⟩

⟨⟨ *The suitability analysis regarding the water sources could be conducted both for domestic water sources (Surface + Ground water) and through also water desalinated imported from the neighbor countries to investigate the water cost implication and guarantee the sustainability of the ground water sources for the different end-users*⟩⟩

⟨⟨ *Knowing the higher solar potential in Niger and the good orographic characteristic of the terrain the site suitability analysis should focus on the economic criterion selection mainly the proximity to the infrastructure and water resources* ⟩⟩

⟨⟨ *In the Suitability analysis the proximity distance to infrastructure such as road, power grid networks and pipeline should be considered with great care, due to the low density of roads and grid regarding the sparsely populated distribution. Focusing on the proximity to the infrastructure criterion weight would discard the other regions with ow density of roads and grid although the huge solar potential while new road could be constructed to decrease the logistic cost* ⟩⟩

⟨⟨ *large-scale solar-based hydrogen production for export required a transport infrastructure, Niger being neighbor countries of Algeria could explore this opportunity to export hydrogen through Algeria to Europe* ⟩⟩

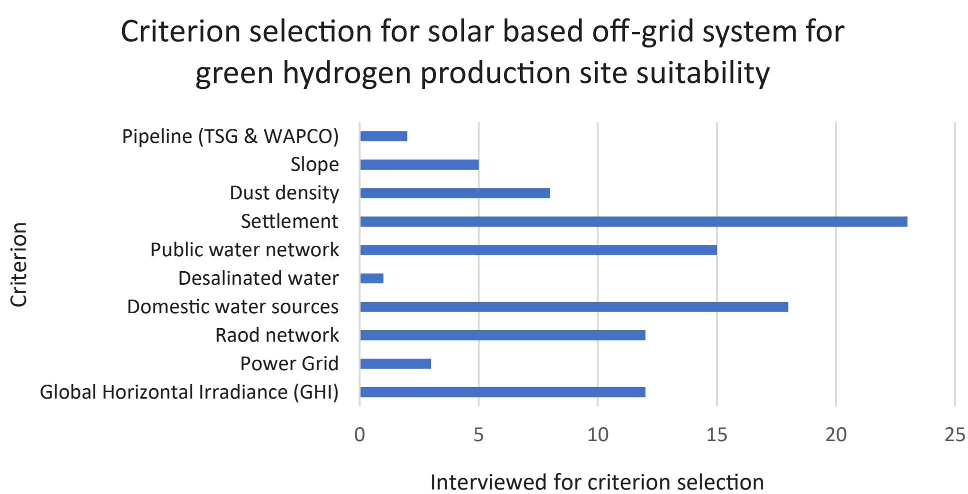

Figure C-1. Land suitability analysis criterion selection survey for off-grid system

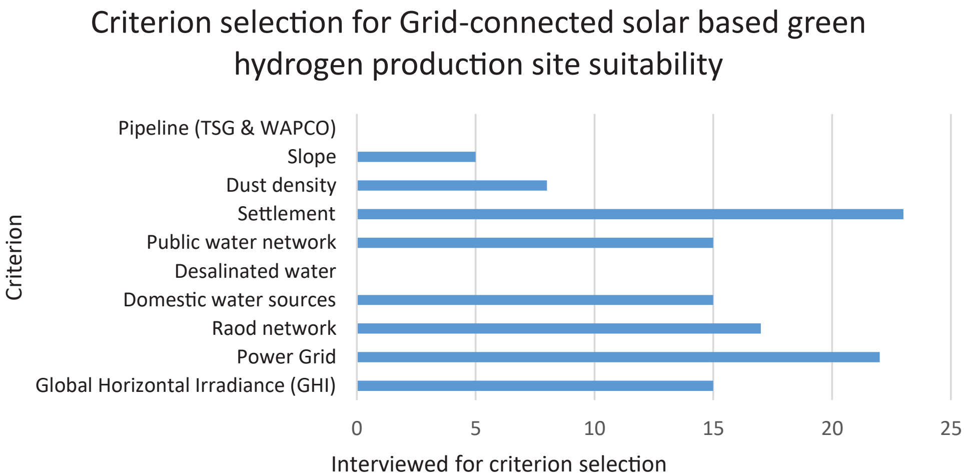

Figure C-2. Land suitability analysis criterion selection survey for grid-connected system

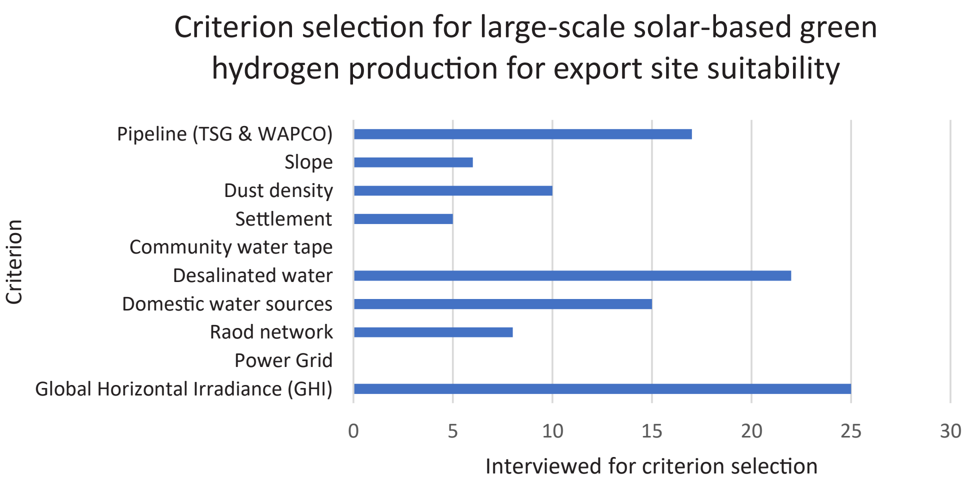

Figure C-3. Land suitability analysis criterion selection survey for large-scale system for export

Data availability

Data will be made available on request.

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